Haokun (Daniel) Xu

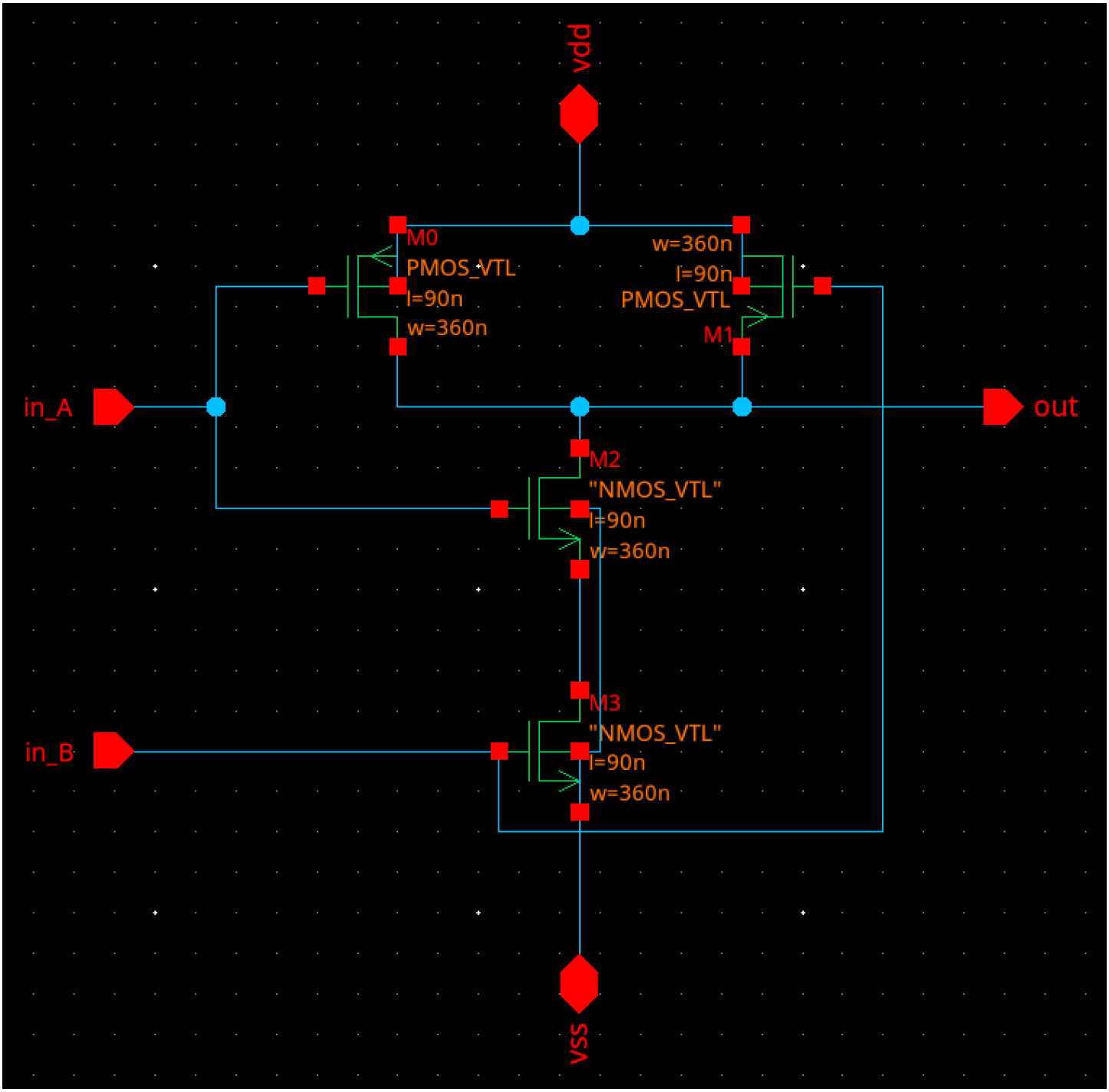
ECE 4385

HW 2

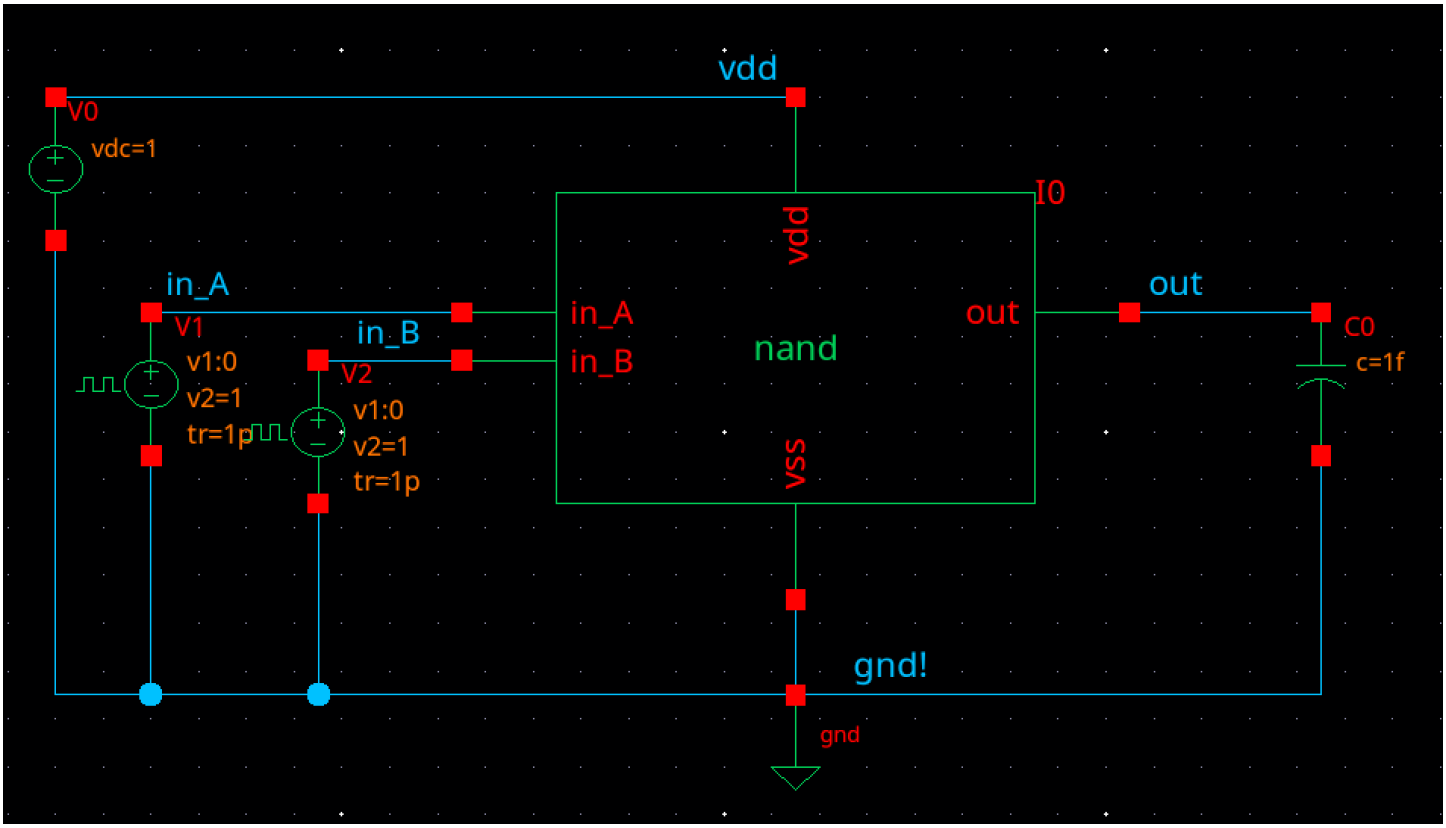
NAND

- I included NAND because Dr. Bhuva talked about including all building blocks of the layout in class today (10/5).

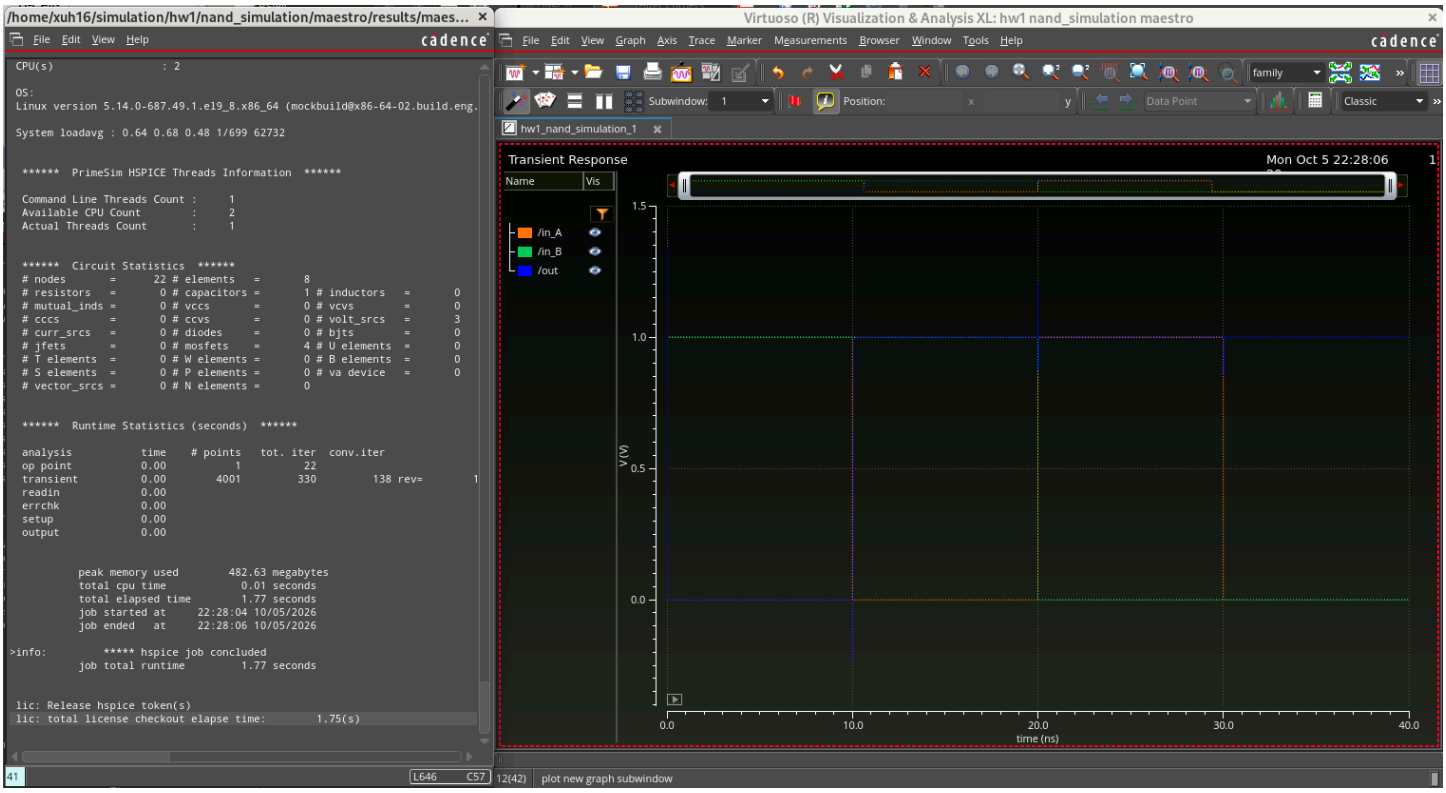
Schematic



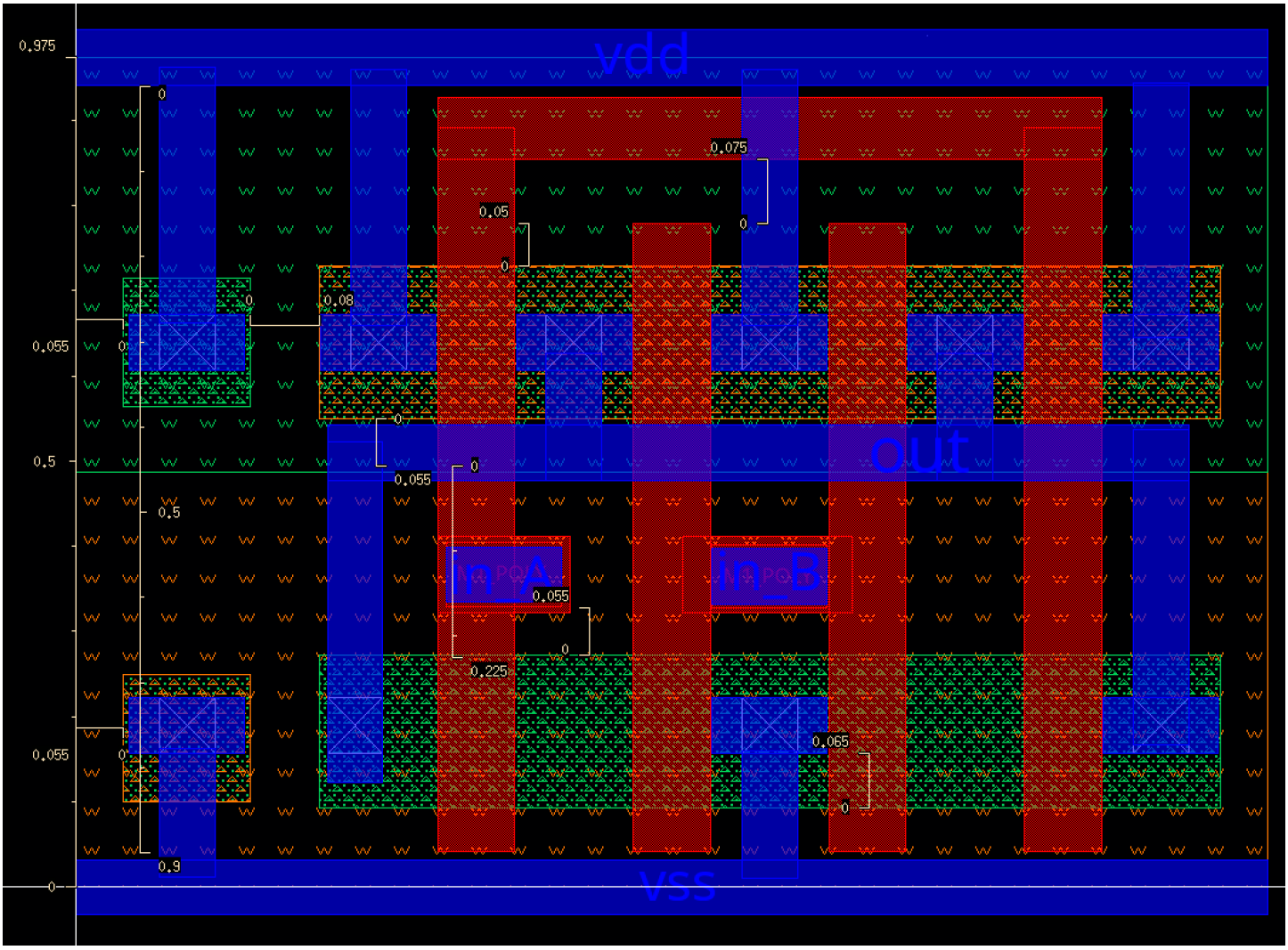
Symbol/Simulation Setup



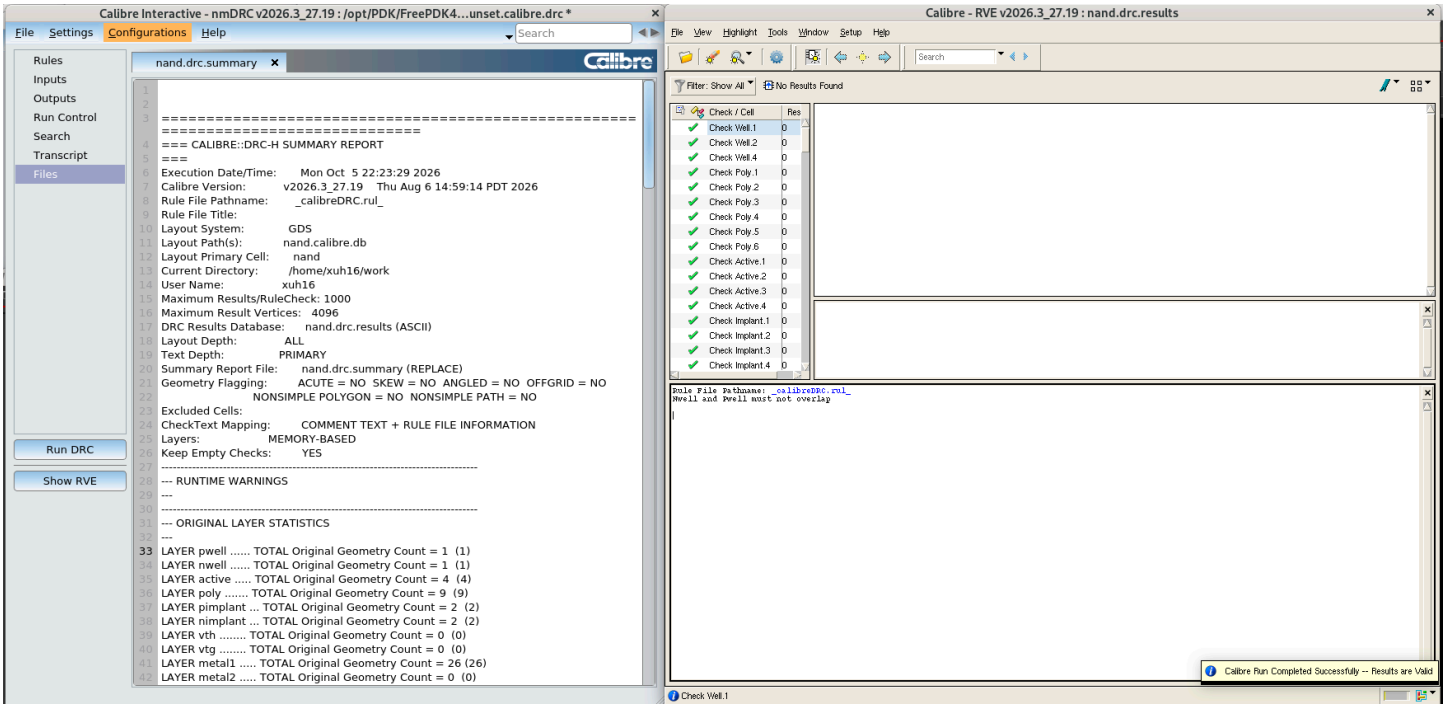
Simulation Results



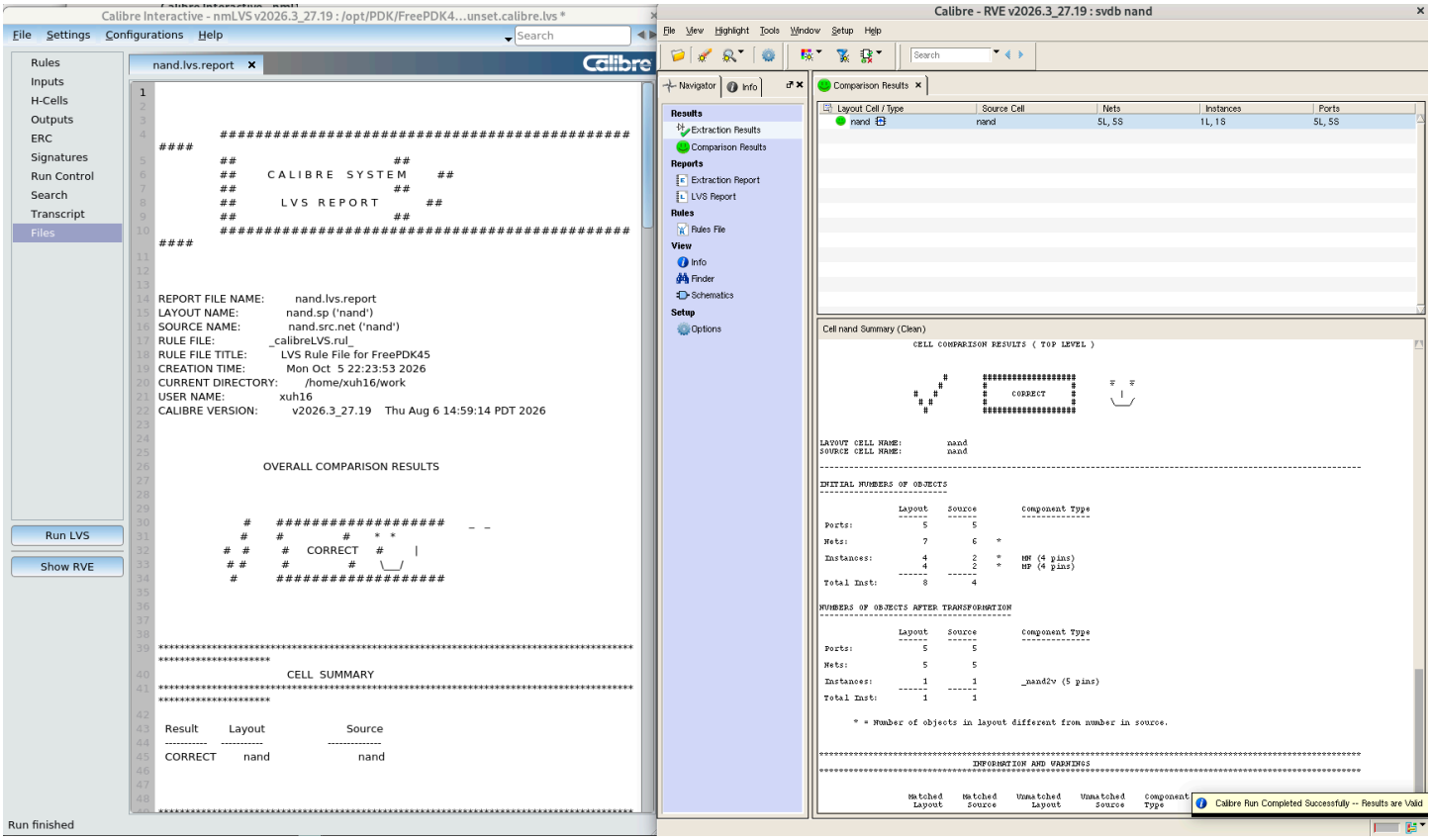
Layout


vdd
out
In_A
In_B
vss
0.975
0.055
0.5
0.055
0
0.9
0.075
0.05
0.08
0.055
0.5
0.055
0.225
0.055

DRC

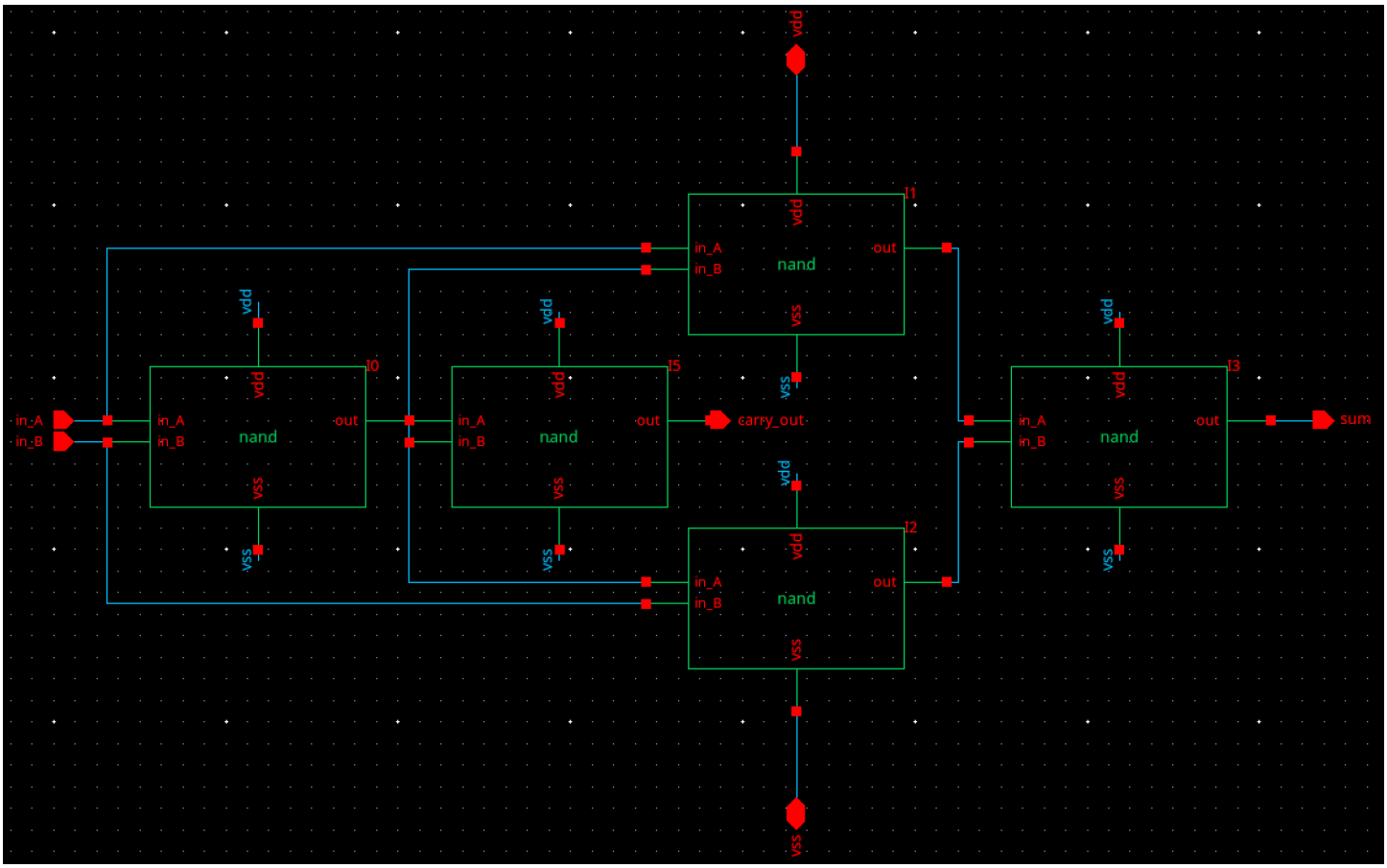

Calibre Interactive - nmDRC v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.drc *
File Settings Configurations Help
Rules
Inputs
Outputs
Run Control
Search
Transcript
Files
Run DRC
Show RVE
nand.drc.summary
=== CALIBRE::DRC-H SUMMARY REPORT
Execution Date/Time: Mon Oct 5 22:23:29 2026
Calibre Version: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
Rule File Pathname: _calibreDRC.rul_
Layout System: GDS
Layout Path(s): nand.calibre.db
Layout Primary Cell: nand
Current Directory: /home/xuh16/work
User Name: xuh16
Maximum Results/RuleCheck: 1000
Maximum Result Vertices: 4096
DRC Results Database: nand.drc.results (ASCII)
Layout Depth: ALL
Text Depth: PRIMARY
Summary Report File: nand.drc.summary (REPLACE)
Geometry Flagging: ACUTE = NO SKEW = NO ANGLED = NO OFFGRID = NO
NONSIMPLE POLYGON = NO NONSIMPLE PATH = NO
Excluded Cells:
CheckText Mapping: COMMENT TEXT + RULE FILE INFORMATION
Layers: MEMORY-BASED
Keep Empty Checks: YES
--- RUNTIME WARNINGS
--- ORIGINAL LAYER STATISTICS
LAYER pwell ..... TOTAL Original Geometry Count = 1 (1)
LAYER nwell ..... TOTAL Original Geometry Count = 1 (1)
LAYER active ..... TOTAL Original Geometry Count = 4 (4)
LAYER poly ..... TOTAL Original Geometry Count = 9 (9)
LAYER pimplant ... TOTAL Original Geometry Count = 2 (2)
LAYER nimplant ... TOTAL Original Geometry Count = 2 (2)
LAYER vth ..... TOTAL Original Geometry Count = 0 (0)
LAYER vtg ..... TOTAL Original Geometry Count = 0 (0)
LAYER metal1 ..... TOTAL Original Geometry Count = 26 (26)
LAYER metal2 ..... TOTAL Original Geometry Count = 0 (0)
Calibre - RVE v2026.3_27.19 : nand.drc.results
File View Highlight Tools Window Setup Help
Filter: Show All
No Results Found
Check / Cell
Check Well.1
Check Well.2
Check Well.4
Check Poly.1
Check Poly.2
Check Poly.3
Check Poly.4
Check Poly.5
Check Poly.6
Check Active.1
Check Active.2
Check Active.3
Check Active.4
Check Implant.1
Check Implant.2
Check Implant.3
Check Implant.4
Rule File Pathname: _calibreDRC.rul_
Nwell and Pwell must not overlap
Calibre Run Completed Successfully -- Results are Valid
Check Well.1

LVS

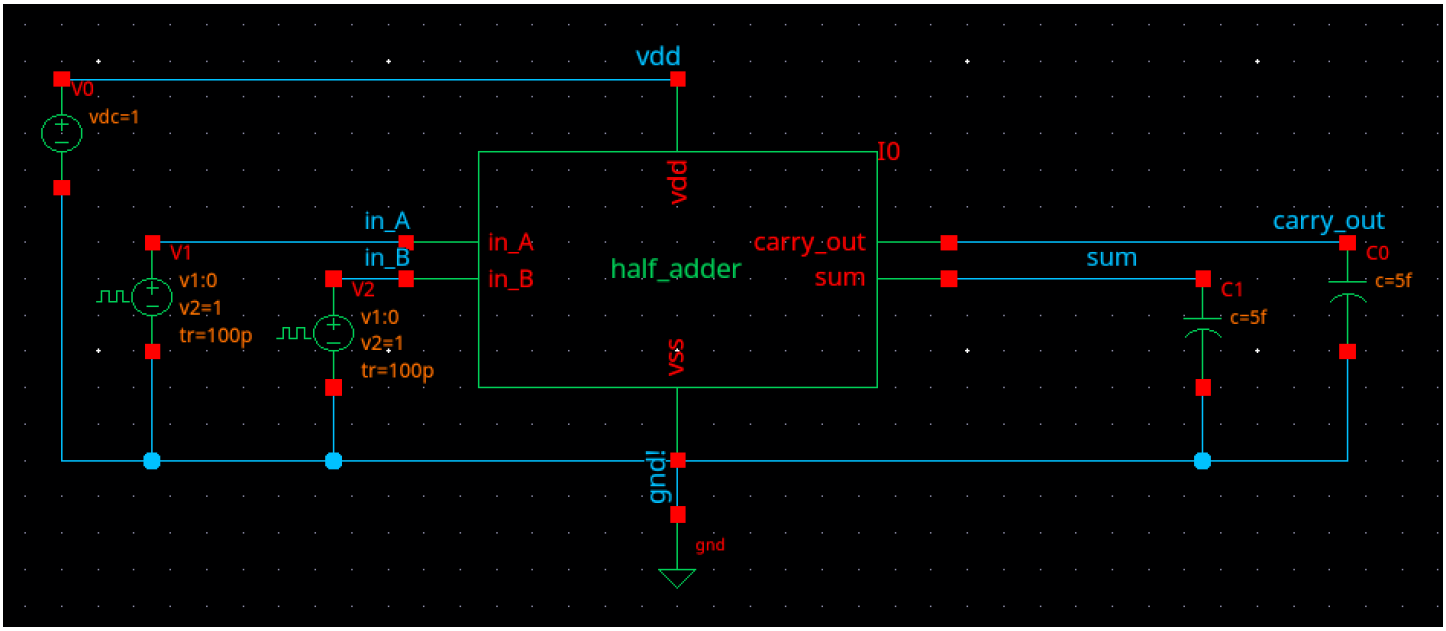

Calibre Interactive - nmLVS v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.lvs *
File Settings Configurations Help
Rules
Inputs
H-Cells
Outputs
ERC
Signatures
Run Control
Search
Transcript
Files
Run LVS
Show RVE
nand.lvs.report
REPORT FILE NAME: nand.lvs.report
LAYOUT NAME: nand.sp ('nand')
SOURCE NAME: nand.src.net ('nand')
RULE FILE: _calibreLVS.rul_
RULE FILE TITLE: LVS Rule File for FreePDK45
CREATION TIME: Mon Oct 5 22:23:53 2026
CURRENT DIRECTORY: /home/xuh16/work
USER NAME: xuh16
CALIBRE VERSION: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
OVERALL COMPARISON RESULTS
CORRECT
CELL SUMMARY
Result Layout Source
CORRECT nand nand
Run finished
Calibre - RVE v2026.3_27.19 : svdb nand
File View Highlight Tools Window Setup Help
Navigator
Info
Results
Extraction Results
Comparison Results
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Comparison Results
Layout Cell / Type
Source Cell
Nets
Instances
Ports
nand
nand
5L, 5S
1L, 1S
5L, 5S
Cell nand Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
LAYOUT CELL NAME: nand
SOURCE CELL NAME: nand
INITIAL NUMBERS OF OBJECTS
Layout Source Component Type
Ports: 5 5
Nets: 7 6 *
Instances: 4 2 * NM (4 pins)
4 2 * MP (4 pins)
Total Inst: 8 4
NUMBERS OF OBJECTS AFTER TRANSFORMATION
Layout Source Component Type
Ports: 5 5
Nets: 5 5
Instances: 1 1 _nand2v (5 pins)
Total Inst: 1 1
* = Number of objects in layout different from number in source.
INFORMATION AND WARNINGS
Matched Layout
Matched Source
Unmatched Layout
Unmatched Source
Component Type
Calibre Run Completed Successfully -- Results are Valid

Half Adder

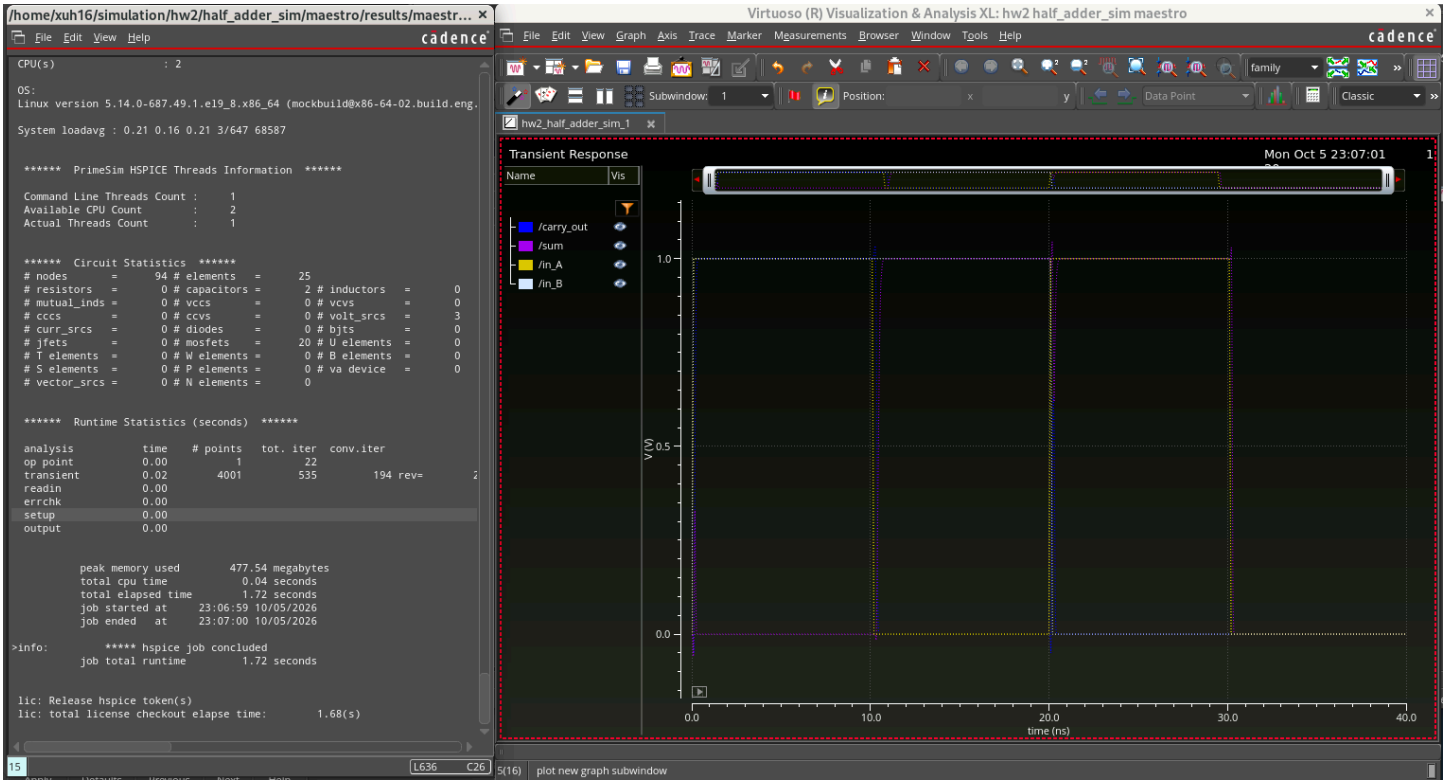
Schematic



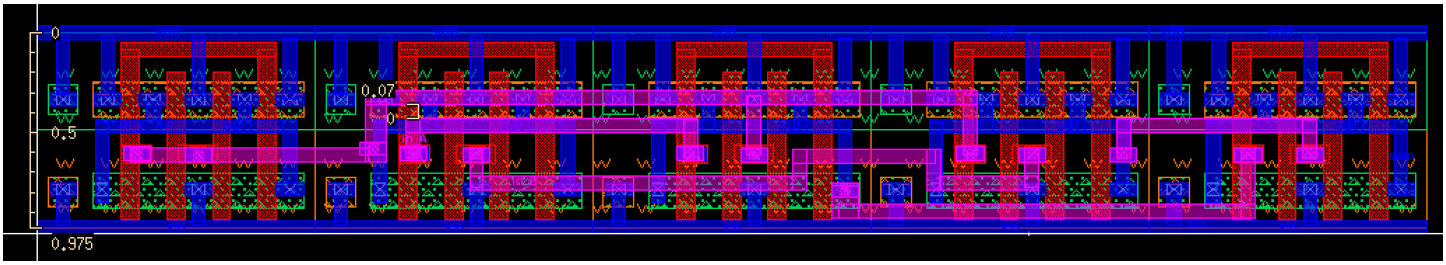
Symbol/Simulation Setup



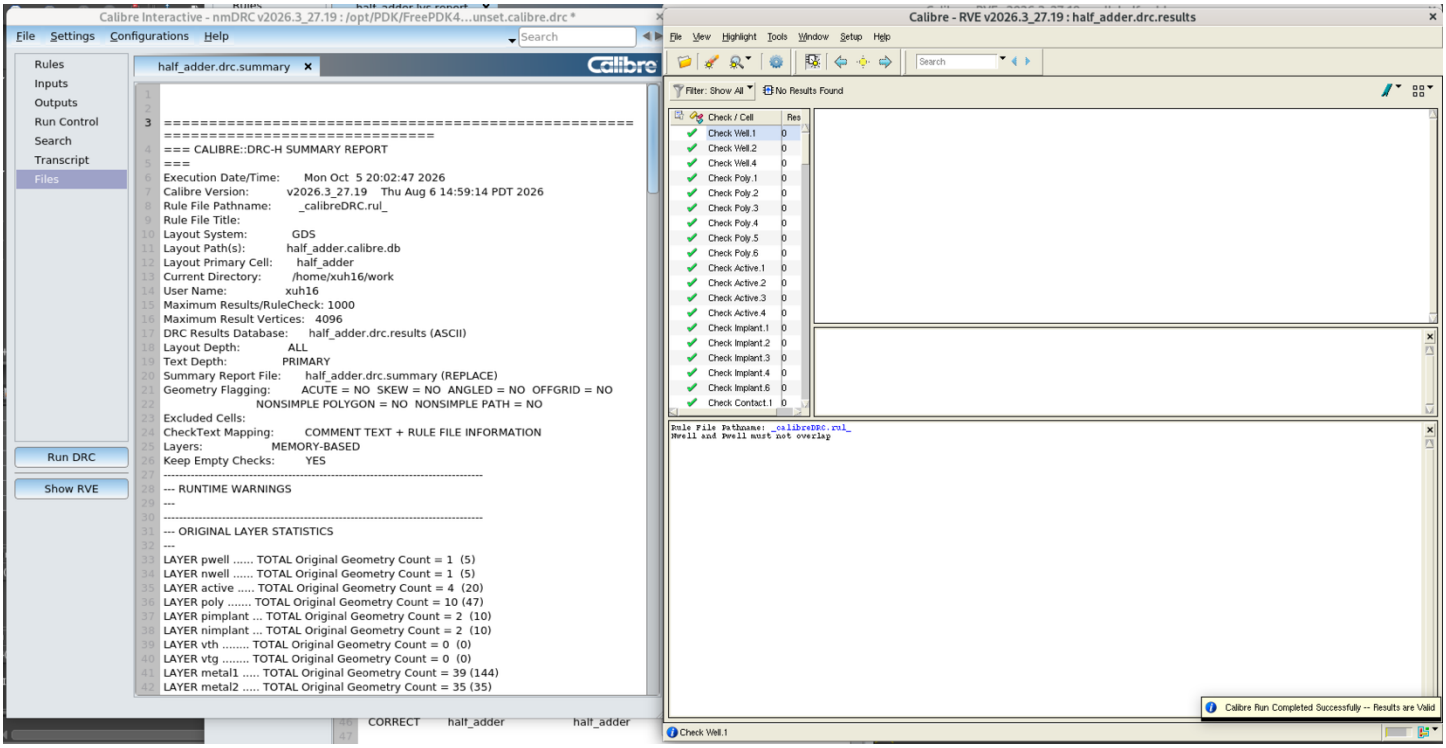
Simulation Results



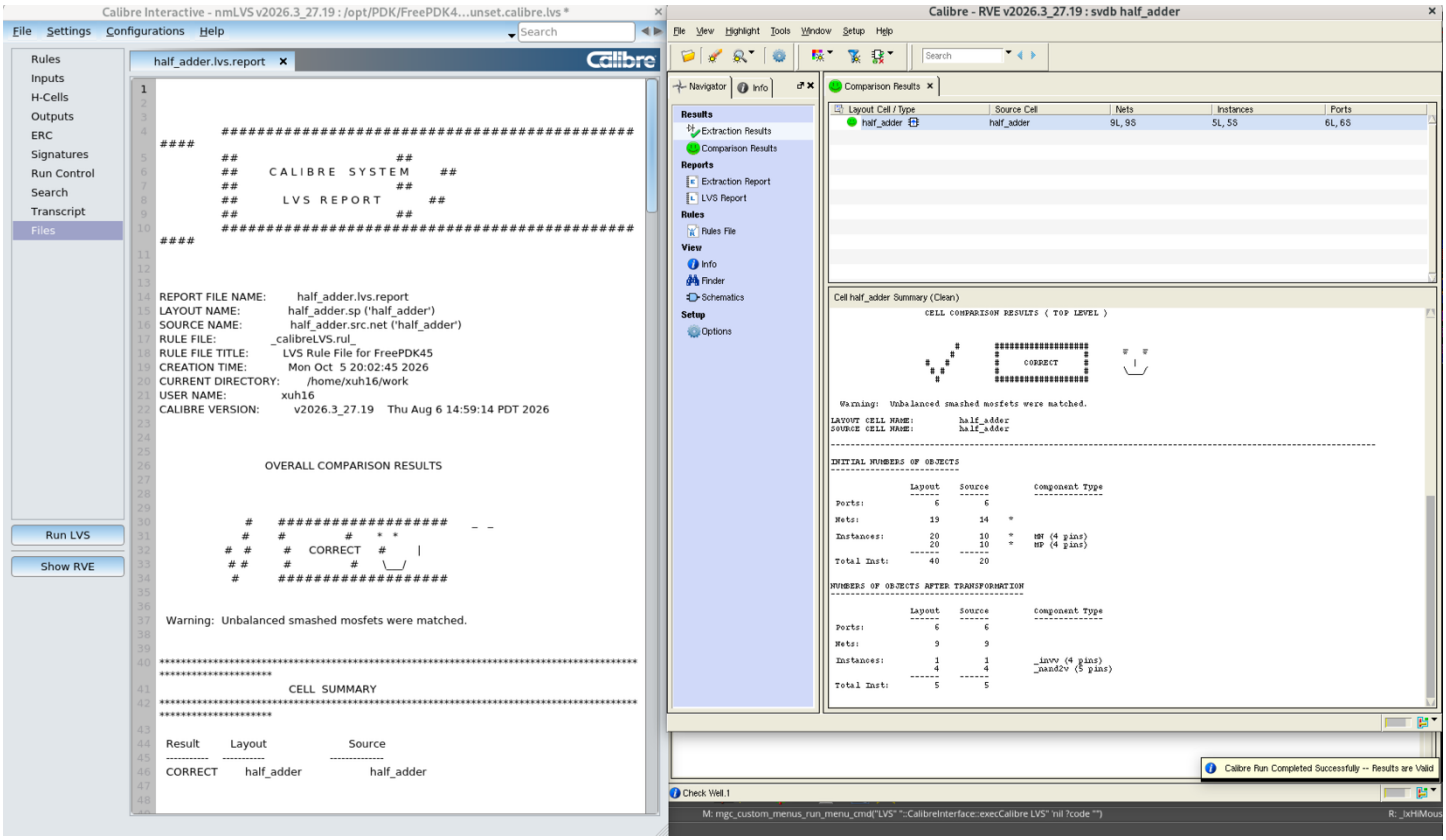
Layout



DRC

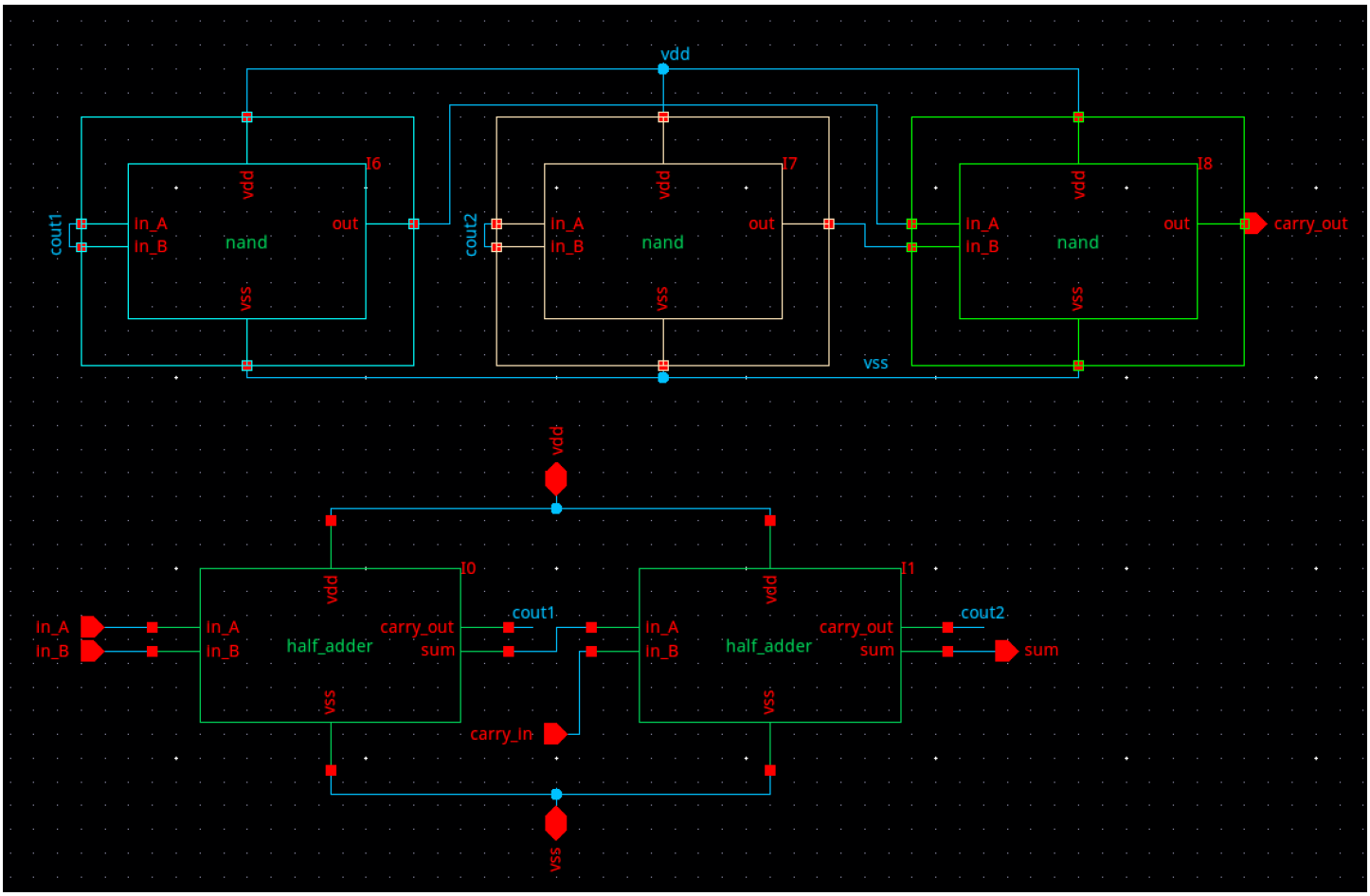

Calibre Interactive - nmDRC v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.drc *
File Settings Configurations Help
Rules
Inputs
Outputs
Run Control
Search
Transcript
Files
Run DRC
Show RVE
half_adder.drc.summary
=== CALIBRE::DRC-H SUMMARY REPORT
Execution Date/Time: Mon Oct 5 20:02:47 2026
Calibre Version: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
Rule File Pathname: _calibreDRC.rul_
Layout System: GDS
Layout Path(s): half_adder.calibre.db
Layout Primary Cell: half_adder
Current Directory: /home/xuh16/work
User Name: xuh16
Maximum Results/RuleCheck: 1000
Maximum Result Vertices: 4096
DRC Results Database: half_adder.drc.results (ASCII)
Layout Depth: ALL
Text Depth: PRIMARY
Summary Report File: half_adder.drc.summary (REPLACE)
Geometry Flagging: ACUTE = NO SKEW = NO ANGLED = NO OFFGRID = NO
NONSIMPLE POLYGON = NO NONSIMPLE PATH = NO
Excluded Cells:
CheckText Mapping: COMMENT TEXT + RULE FILE INFORMATION
Layers: MEMORY-BASED
Keep Empty Checks: YES
--- RUNTIME WARNINGS
--- ORIGINAL LAYER STATISTICS
LAYER pwell ..... TOTAL Original Geometry Count = 1 (5)
LAYER nwell ..... TOTAL Original Geometry Count = 1 (5)
LAYER active ..... TOTAL Original Geometry Count = 4 (20)
LAYER poly ..... TOTAL Original Geometry Count = 10 (47)
LAYER pimplant ... TOTAL Original Geometry Count = 2 (10)
LAYER nimplant ... TOTAL Original Geometry Count = 2 (10)
LAYER vth ..... TOTAL Original Geometry Count = 0 (0)
LAYER vtg ..... TOTAL Original Geometry Count = 0 (0)
LAYER metal1 ..... TOTAL Original Geometry Count = 39 (144)
LAYER metal2 ..... TOTAL Original Geometry Count = 35 (35)
CORRECT half_adder half_adder
Calibre - RVE v2026.3_27.19 : half_adder.drc.results
File View Highlight Tools Window Setup Help
Filter: Show All
No Results Found
Check / Cell
Check Well.1
Check Well.2
Check Well.4
Check Poly.1
Check Poly.2
Check Poly.3
Check Poly.4
Check Poly.5
Check Poly.6
Check Active.1
Check Active.2
Check Active.3
Check Active.4
Check Implant.1
Check Implant.2
Check Implant.3
Check Implant.4
Check Implant.6
Check Contact.1
Rule File Pathname: _calibreDRC.rul_
Nwell and Pwell must not overlap
Calibre Run Completed Successfully -- Results are Valid
Check Well.1

LVS

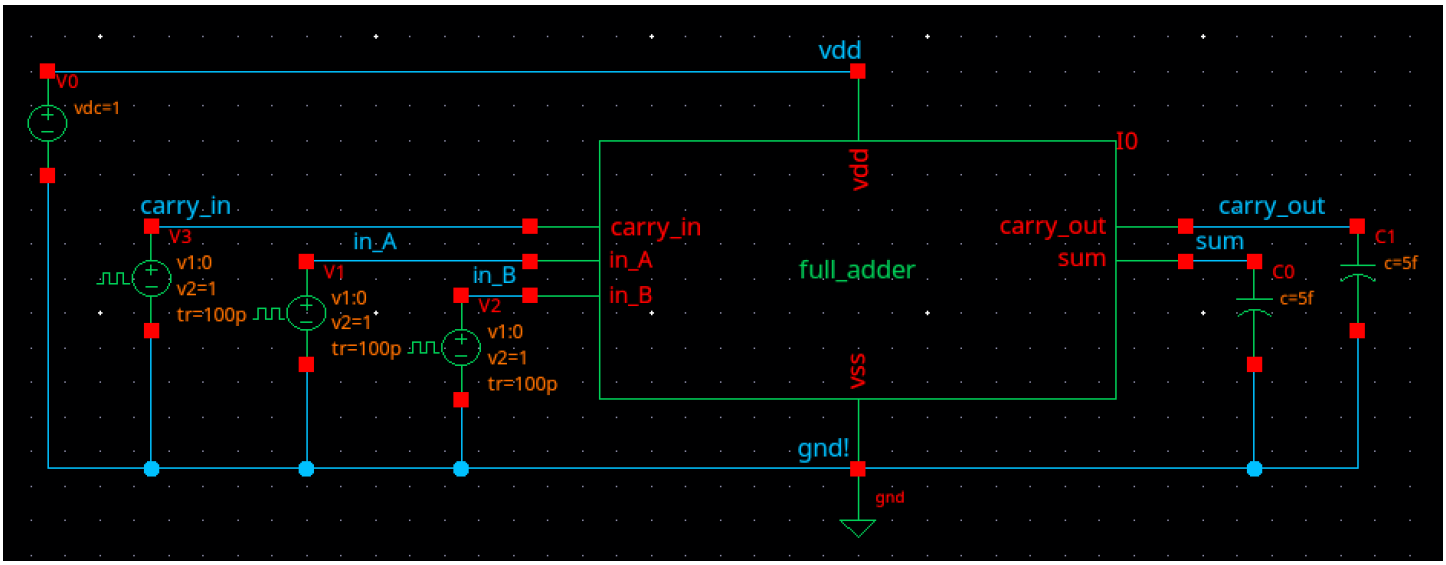

Calibre Interactive - nmLVS v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.lvs *
File Settings Configurations Help
Rules
Inputs
H-Cells
Outputs
ERC
Signatures
Run Control
Search
Transcript
Files
Run LVS
Show RVE
half_adder.lvs.report
CALIBRE SYSTEM
LVS REPORT
REPORT FILE NAME: half_adder.lvs.report
LAYOUT NAME: half_adder.sp ('half_adder')
SOURCE NAME: half_adder.src.net ('half_adder')
RULE FILE: _calibreLVS.rul_
RULE FILE TITLE: LVS Rule File for FreePDK45
CREATION TIME: Mon Oct 5 20:02:45 2026
CURRENT DIRECTORY: /home/xuh16/work
USER NAME: xuh16
CALIBRE VERSION: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
OVERALL COMPARISON RESULTS
CORRECT
Warning: Unbalanced smashed mosfets were matched.
CELL SUMMARY
Result Layout Source
CORRECT half_adder half_adder
Calibre - RVE v2026.3_27.19 : svdb half_adder
Navigator
Info
Comparison Results
Results
Extraction Results
Comparison Results
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Layout Cell / Type
Source Cell
Nets
Instances
Ports
half_adder
Cell half_adder Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
Warning: Unbalanced smashed mosfets were matched.
LAYOUT CELL NAME: half_adder
SOURCE CELL NAME: half_adder
INITIAL NUMBERS OF OBJECTS
Ports: 6 6
Nets: 19 14 *
Instances: 20 10 * NM (4 pins)
20 10 * MP (4 pins)
Total Inst: 40 20
NUMBERS OF OBJECTS AFTER TRANSFORMATION
Ports: 6 6
Nets: 9 9
Instances: 1 1 _invv (4 pins)
4 4 _nand2v (5 pins)
Total Inst: 5 5
Calibre Run Completed Successfully -- Results are Valid
Check Well.1

Full Adder

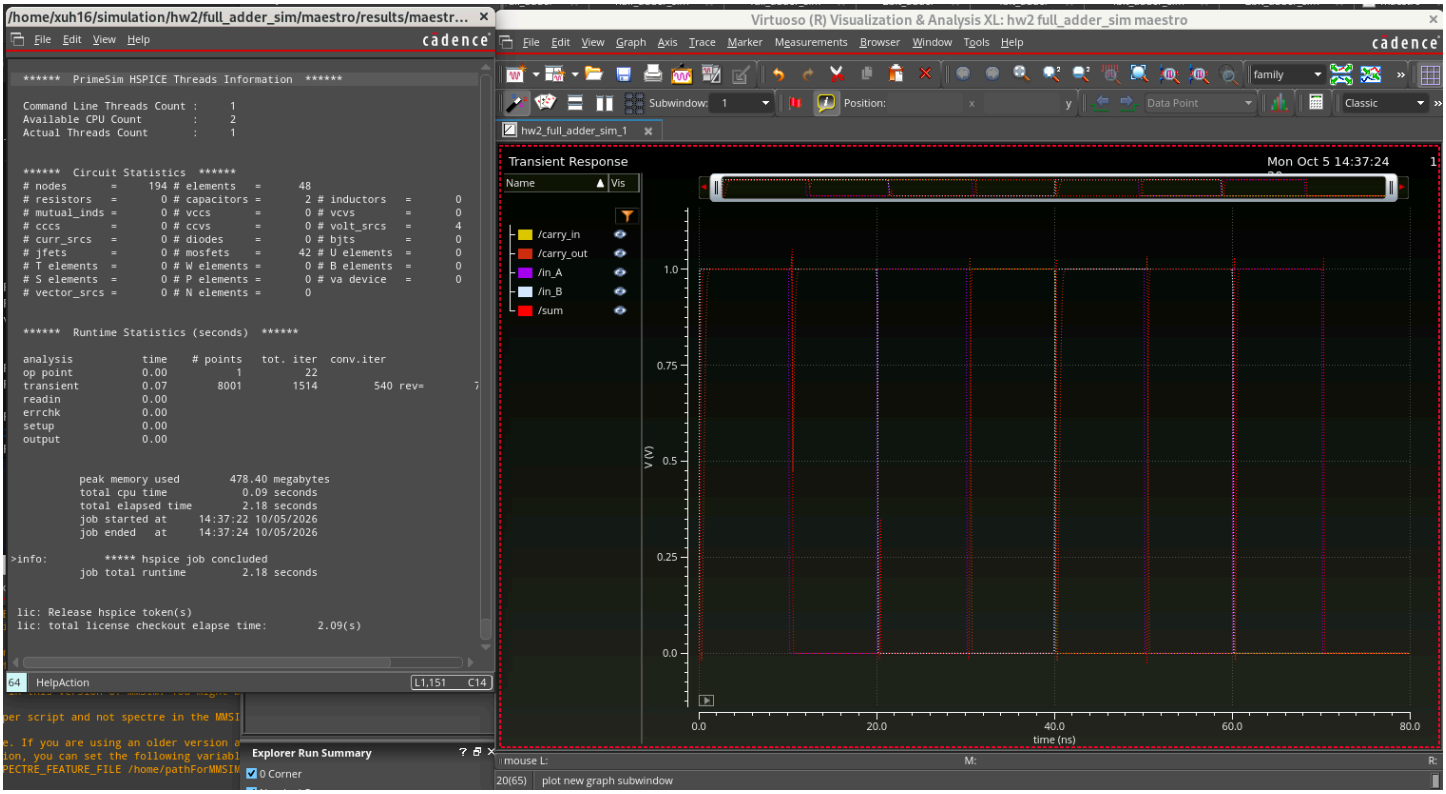
Schematic



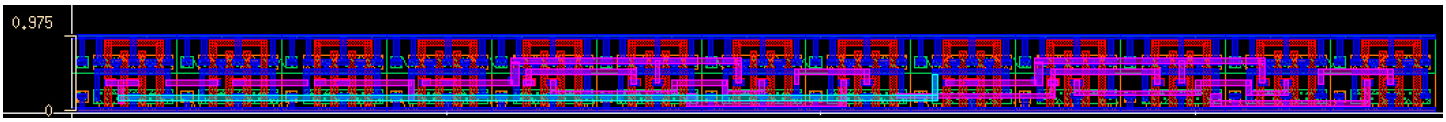
Symbol/Simulation Setup



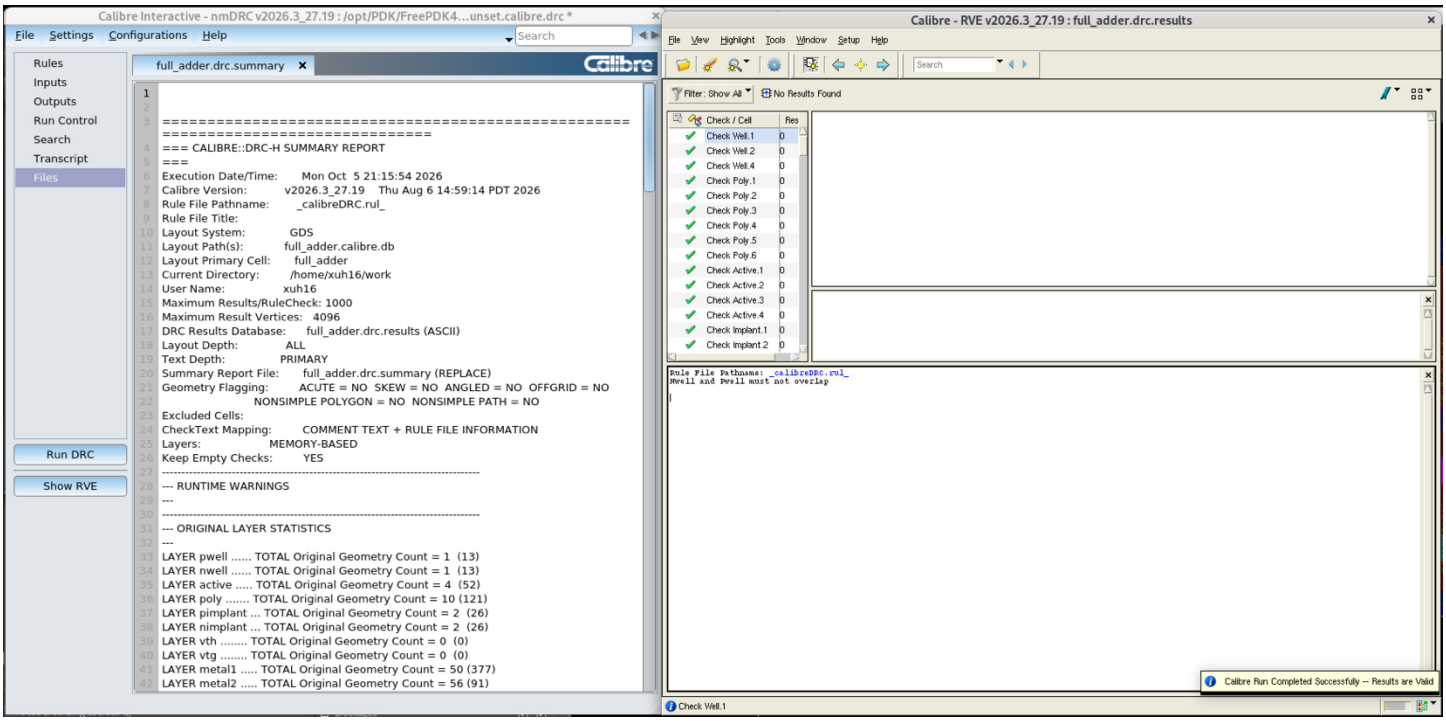
Simulation Results



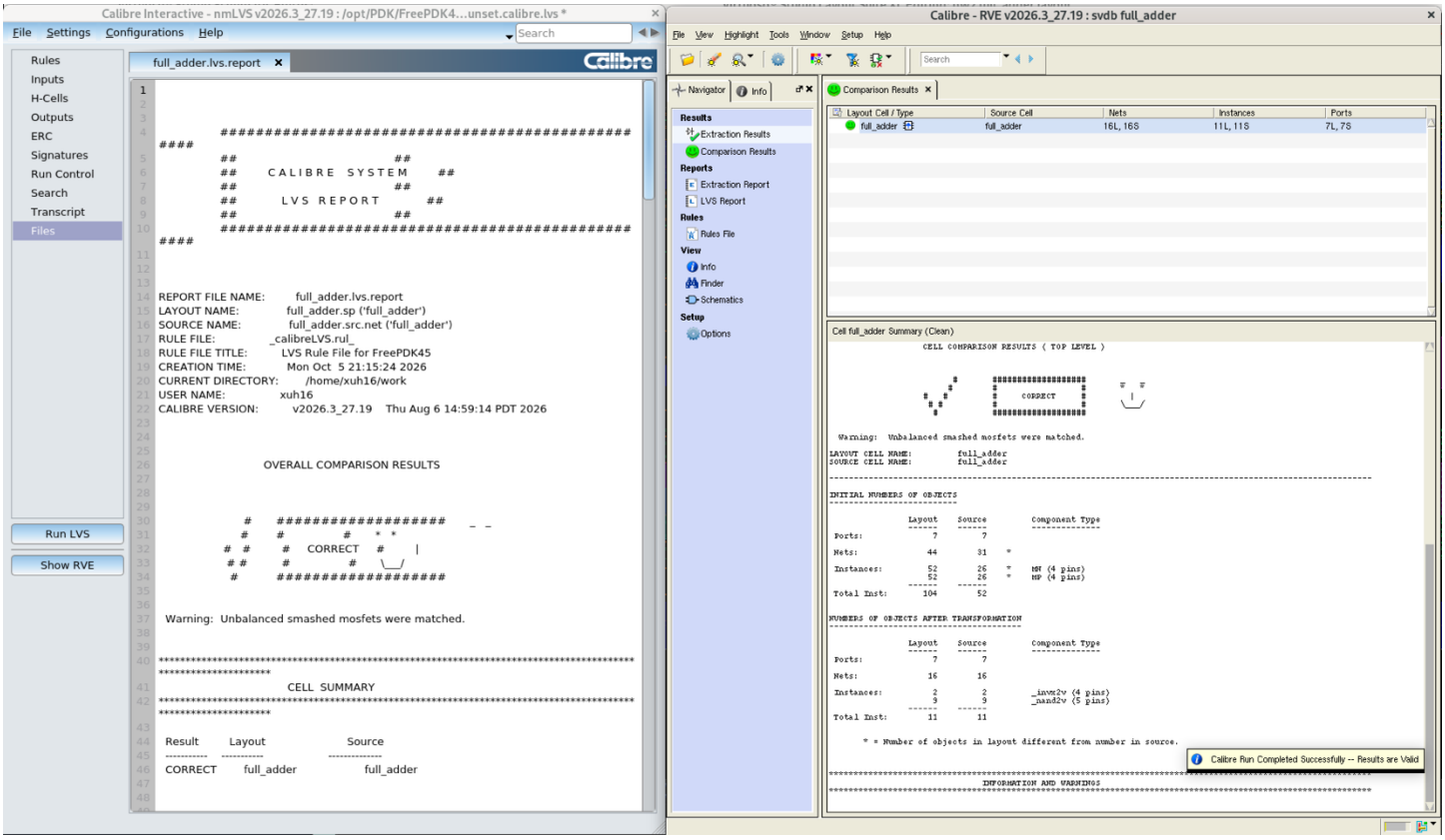
Layout



DRC

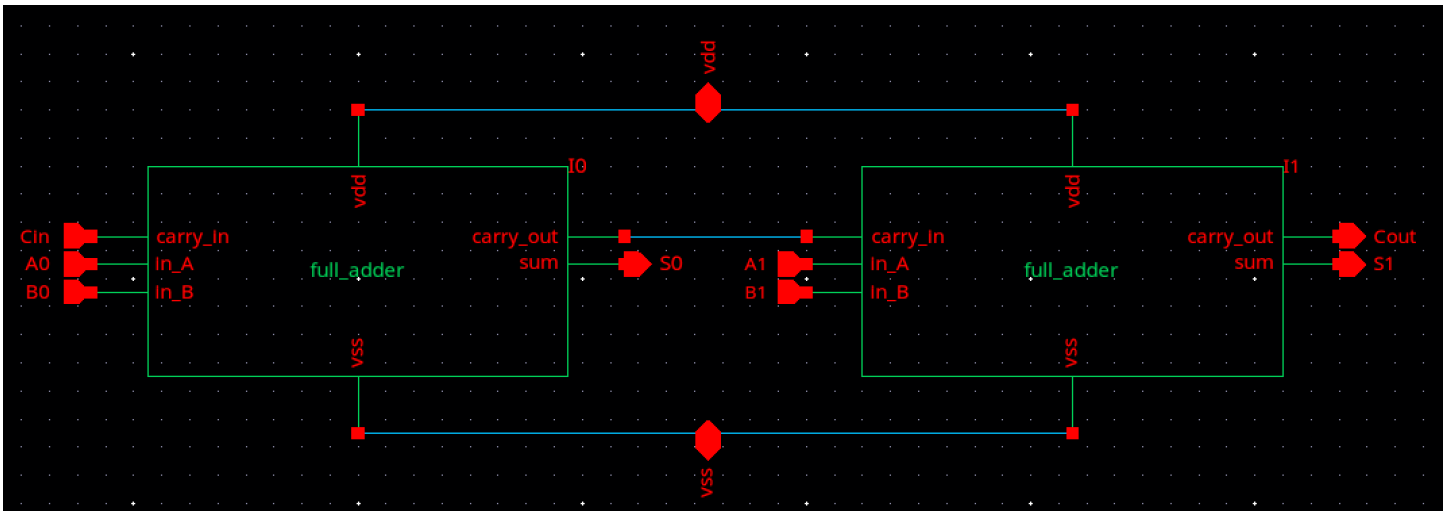


LVS

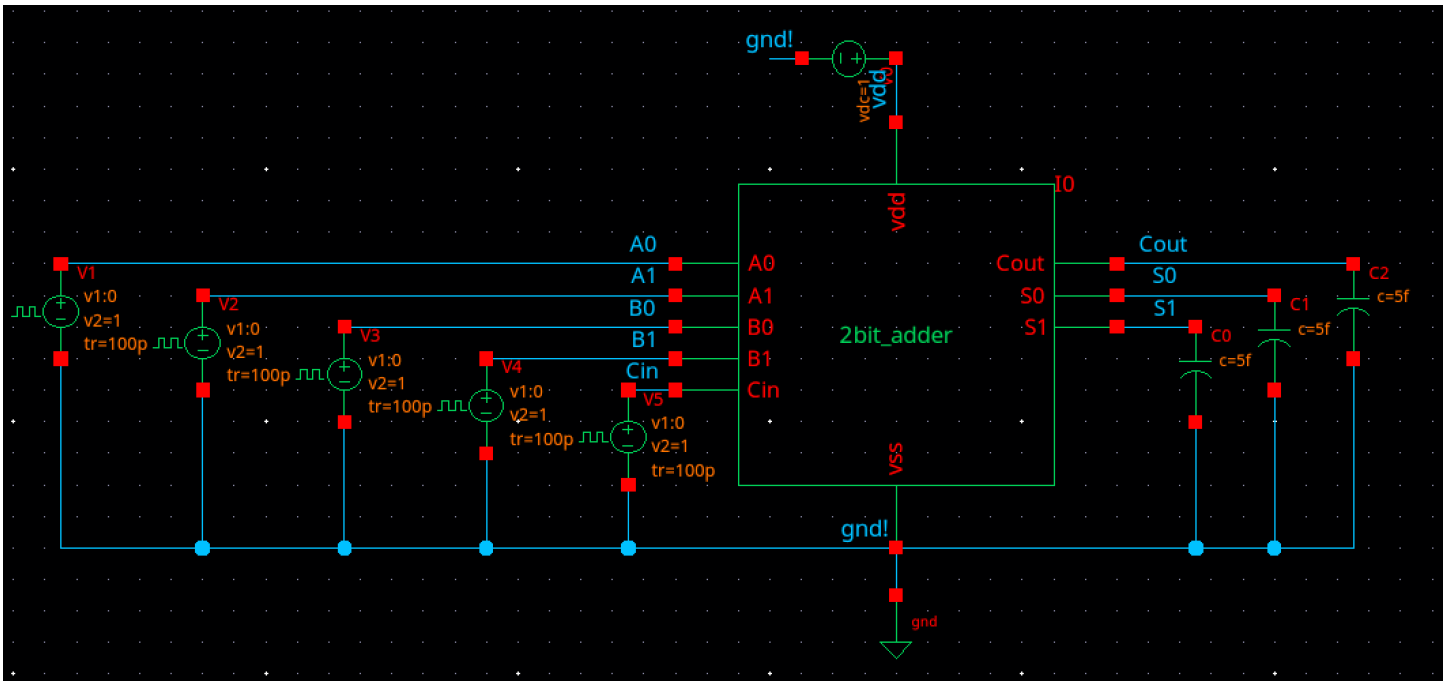

Calibre Interactive - nmLVS v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.lvs *
File Settings Configurations Help
Search
Rules
Inputs
H-Cells
Outputs
ERC
Signatures
Run Control
Search
Transcript
Files
Run LVS
Show RVE
full_adder.lvs.report
CALIBRE SYSTEM
LVS REPORT
REPORT FILE NAME: full_adder.lvs.report
LAYOUT NAME: full_adder.sp ('full_adder')
SOURCE NAME: full_adder.src.net ('full_adder')
RULE FILE: _calibreLVS.rul_
RULE FILE TITLE: LVS Rule File for FreePDK45
CREATION TIME: Mon Oct 5 21:15:24 2026
CURRENT DIRECTORY: /home/xuh16/work
USER NAME: xuh16
CALIBRE VERSION: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
OVERALL COMPARISON RESULTS
CORRECT
Warning: Unbalanced smashed mosfets were matched.
CELL SUMMARY
Result Layout Source
CORRECT full_adder full_adder
Calibre - RVE v2026.3_27.19 : svdb full_adder
File View Highlight Tools Window Setup Help
Navigator
Info
Results
Extraction Results
Comparison Results
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Comparison Results
Layout Cell / Type
Source Cell
Nets
Instances
Ports
full_adder
full_adder
16L, 16S
11L, 11S
7L, 7S
Cell full_adder Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
Warning: Unbalanced smashed mosfets were matched.
LAYOUT CELL NAME: full_adder
SOURCE CELL NAME: full_adder
INITIAL NUMBERS OF OBJECTS
Layout Source Component Type
Ports: 7 7
Nets: 44 31 *
Instances: 52 26 * MM (4 pins)
52 26 * MP (4 pins)
Total Inst: 104 52
NUMBERS OF OBJECTS AFTER TRANSFORMATION
Layout Source Component Type
Ports: 7 7
Nets: 16 16
Instances: 2 2 _invx2v (4 pins)
9 9 _nand2v (5 pins)
Total Inst: 11 11
* = Number of objects in layout different from number in source.
Calibre Run Completed Successfully -- Results are Valid
INFORMATION AND WARNINGS

2-Bit Adder

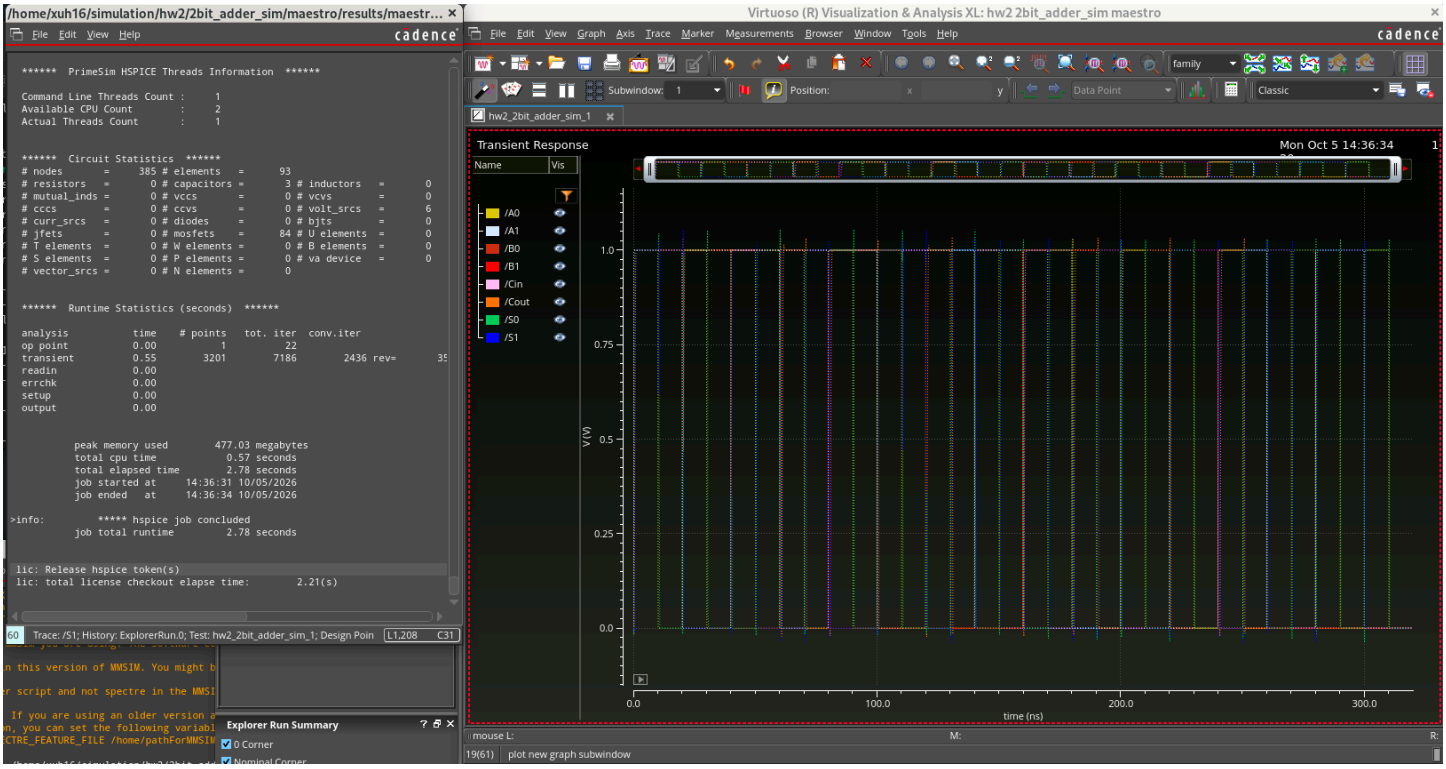
Schematic



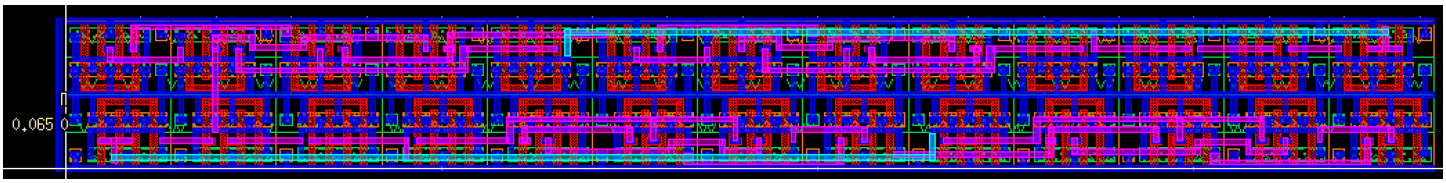
Symbol/Simulation Setup



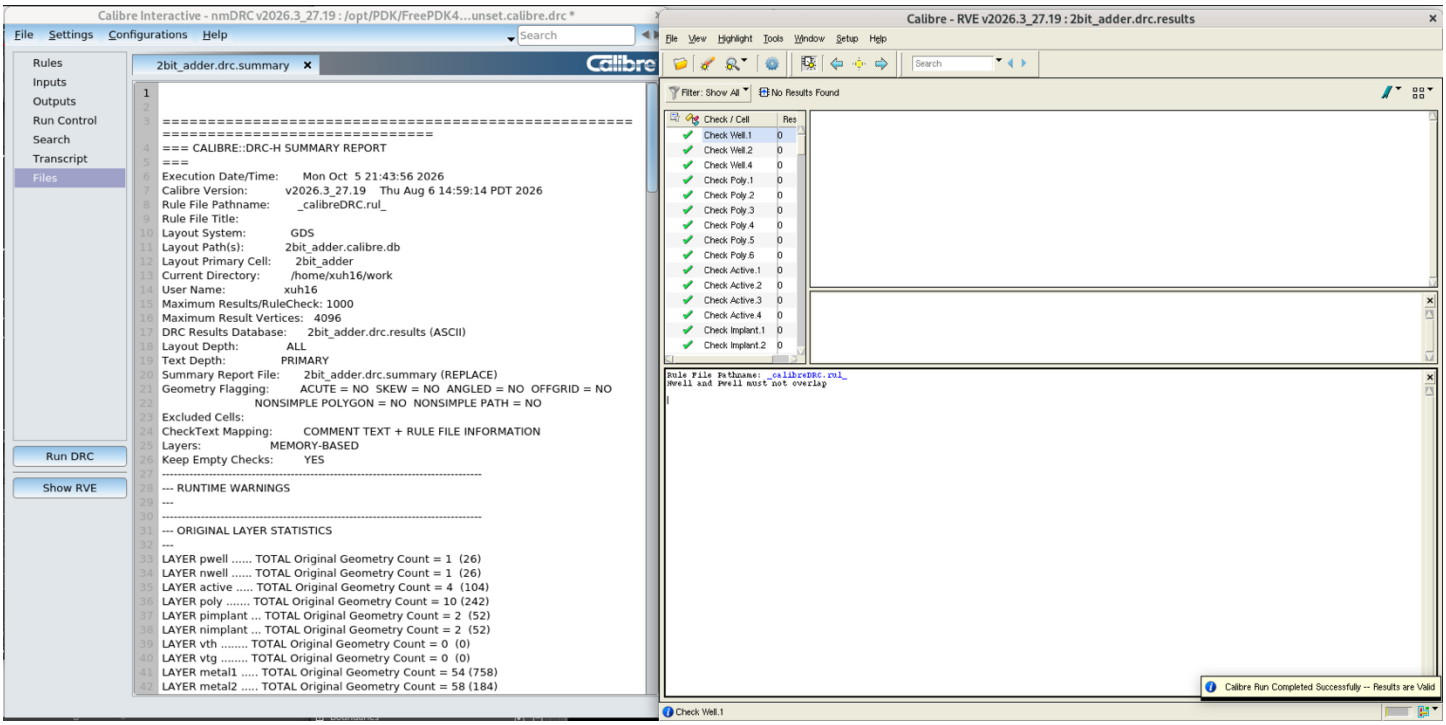
Simulation Results



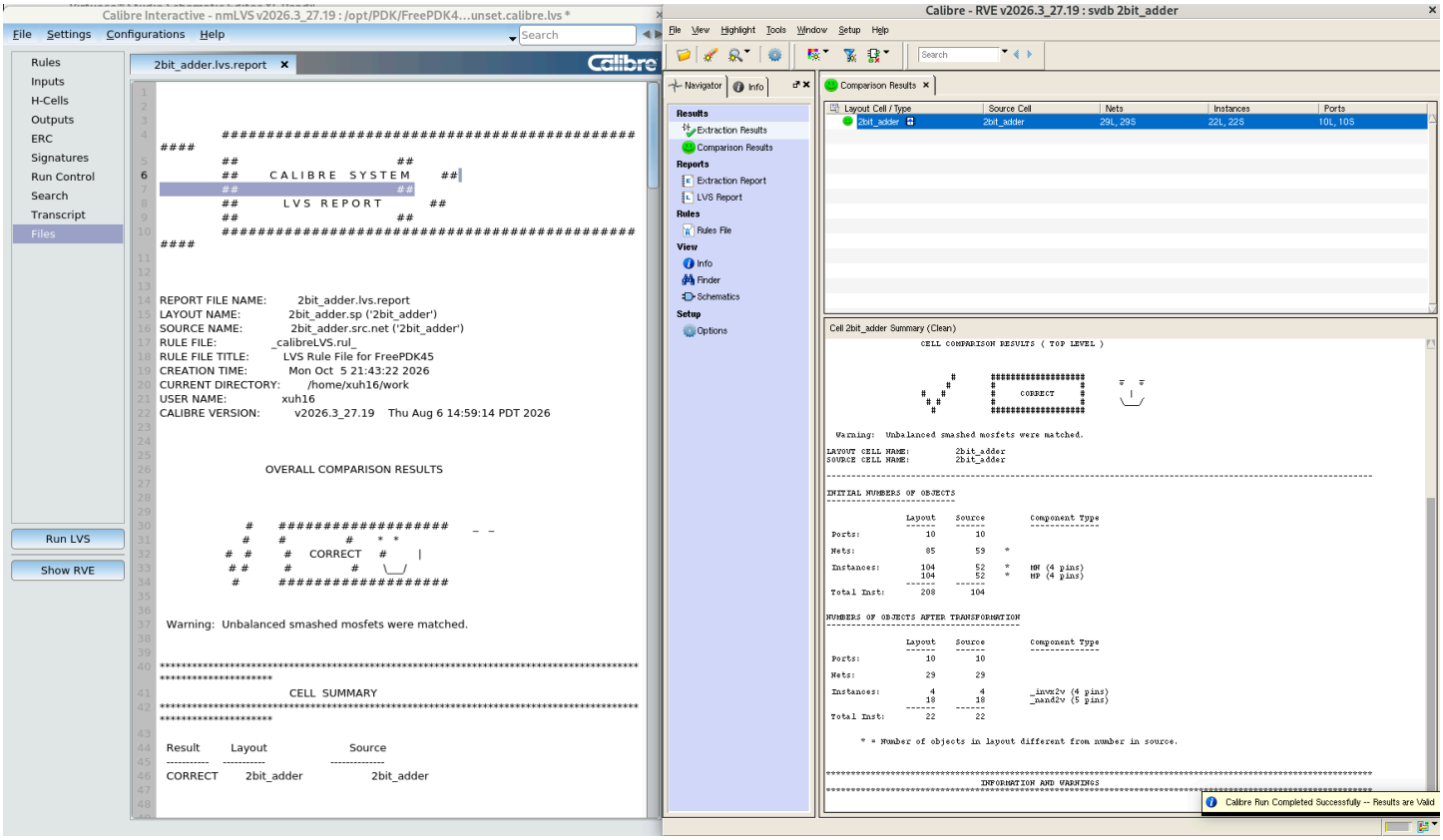
Layout



DRC

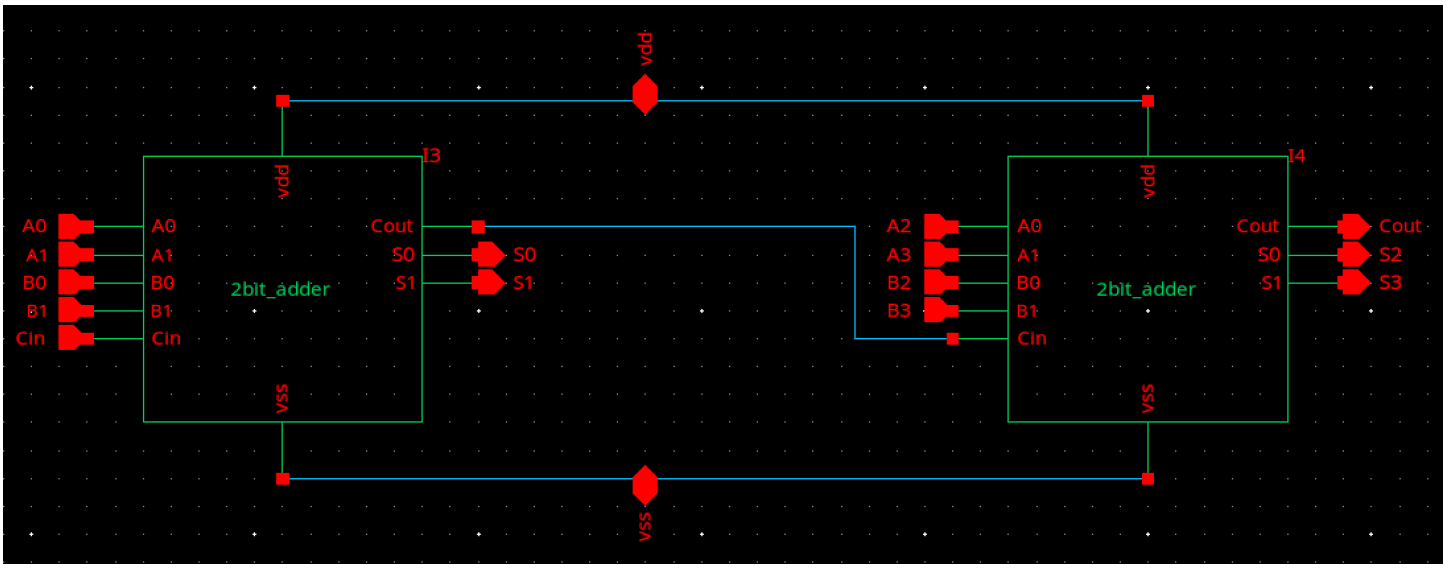


LVS

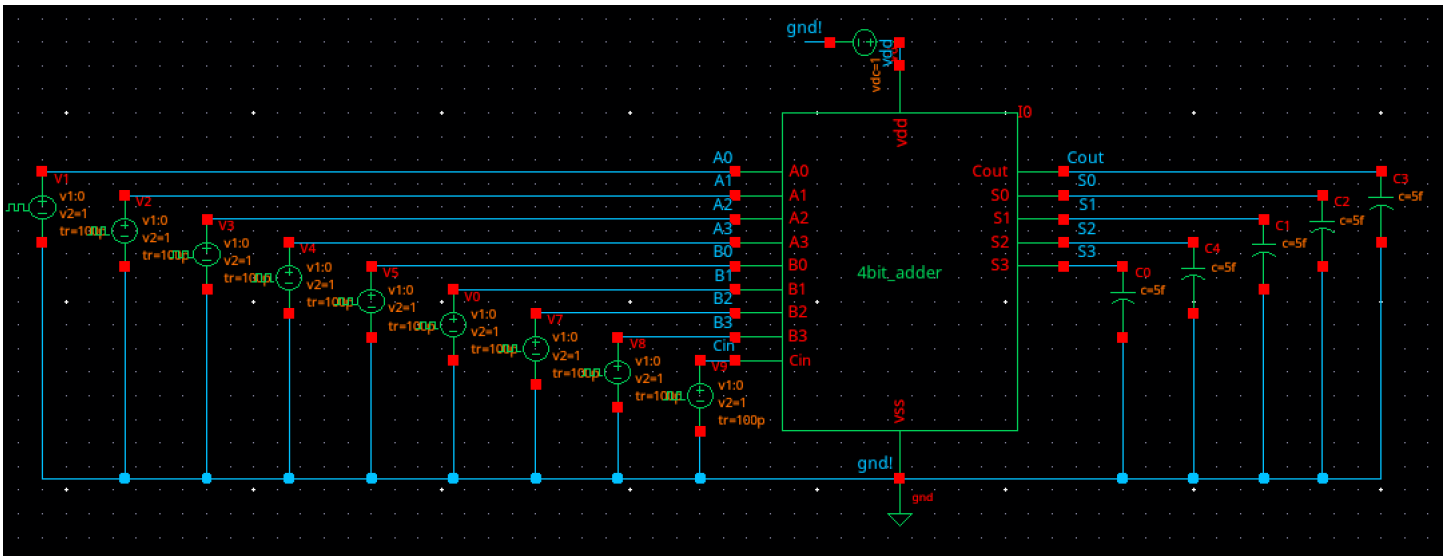

Calibre Interactive - nmLVS v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.lvs *
File Settings Configurations Help
Rules
Inputs
H-Cells
Outputs
ERC
Signatures
Run Control
Search
Transcript
Files
Run LVS
Show RVE
2bit_adder.lvs.report
CALIBRE SYSTEM
LVS REPORT
REPORT FILE NAME: 2bit_adder.lvs.report
LAYOUT NAME: 2bit_adder.sp ('2bit_adder')
SOURCE NAME: 2bit_adder.src.net ('2bit_adder')
RULE FILE: _calibreLVS.rul_
RULE FILE TITLE: LVS Rule File for FreePDK45
CREATION TIME: Mon Oct 5 21:43:22 2026
CURRENT DIRECTORY: /home/xuh16/work
USER NAME: xuh16
CALIBRE VERSION: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
OVERALL COMPARISON RESULTS
CORRECT
Warning: Unbalanced smashed mosfets were matched.
CELL SUMMARY
Result Layout Source
CORRECT 2bit_adder 2bit_adder
Calibre - RVE v2026.3_27.19 : svdb 2bit_adder
File View Highlight Tools Window Setup Help
Navigator
Info
Results
Extraction Results
Comparison Results
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Comparison Results
Layout Cell / Type
Source Cell
Nets
Instances
Ports
2bit_adder
2bit_adder
29L, 29S
22L, 22S
10L, 10S
Cell 2bit_adder Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
Warning: Unbalanced smashed mosfets were matched.
LAYOUT CELL NAME: 2bit_adder
SOURCE CELL NAME: 2bit_adder
INITIAL NUMBERS OF OBJECTS
Layout Source Component Type
Ports: 10 10
Nets: 85 59 *
Instances: 104 52 * NM (4 pins)
104 52 * MP (4 pins)
Total Inst: 208 104
NUMBERS OF OBJECTS AFTER TRANSFORMATION
Layout Source Component Type
Ports: 10 10
Nets: 29 29
Instances: 4 4 _invx2v (4 pins)
18 18 _nand2v (5 pins)
Total Inst: 22 22
* = Number of objects in layout different from number in source.
INFORMATION AND WARNINGS
Calibre Run Completed Successfully -- Results are Valid

4-Bit Adder

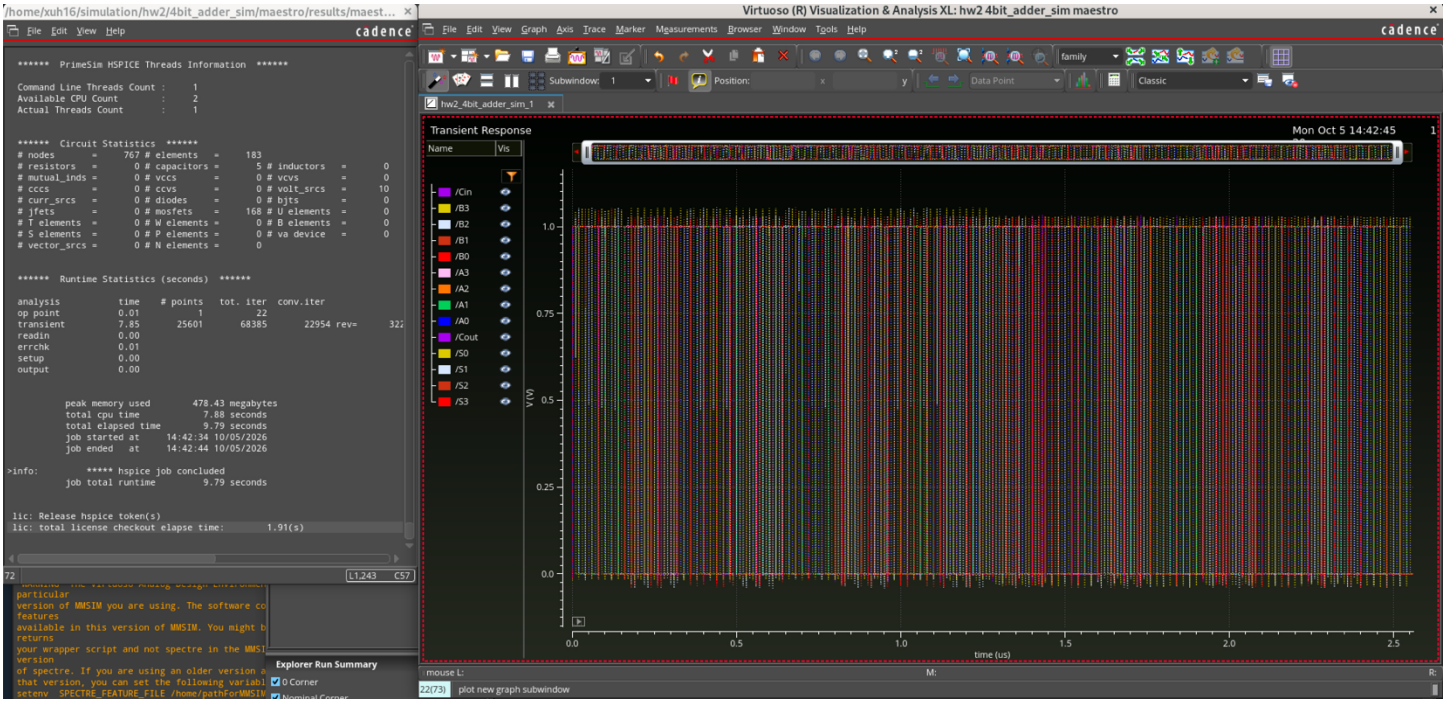
Schematic



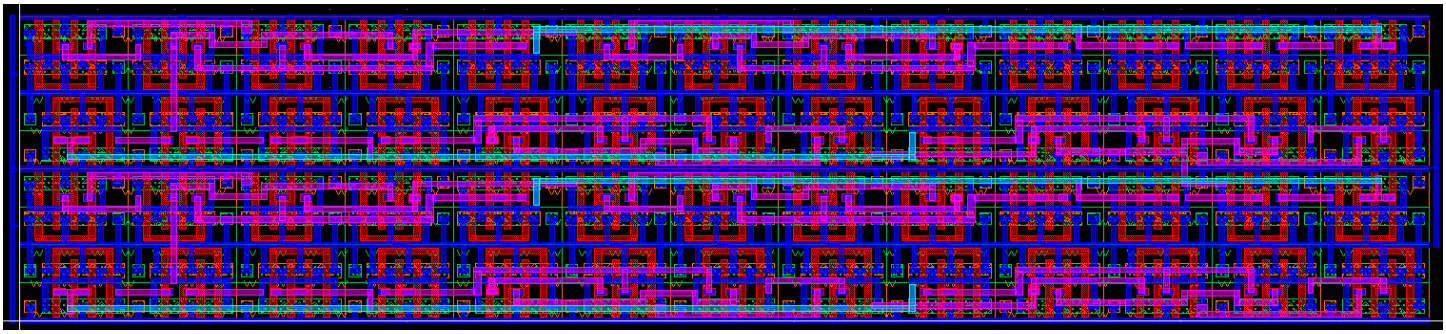
Symbol/Simulation Setup



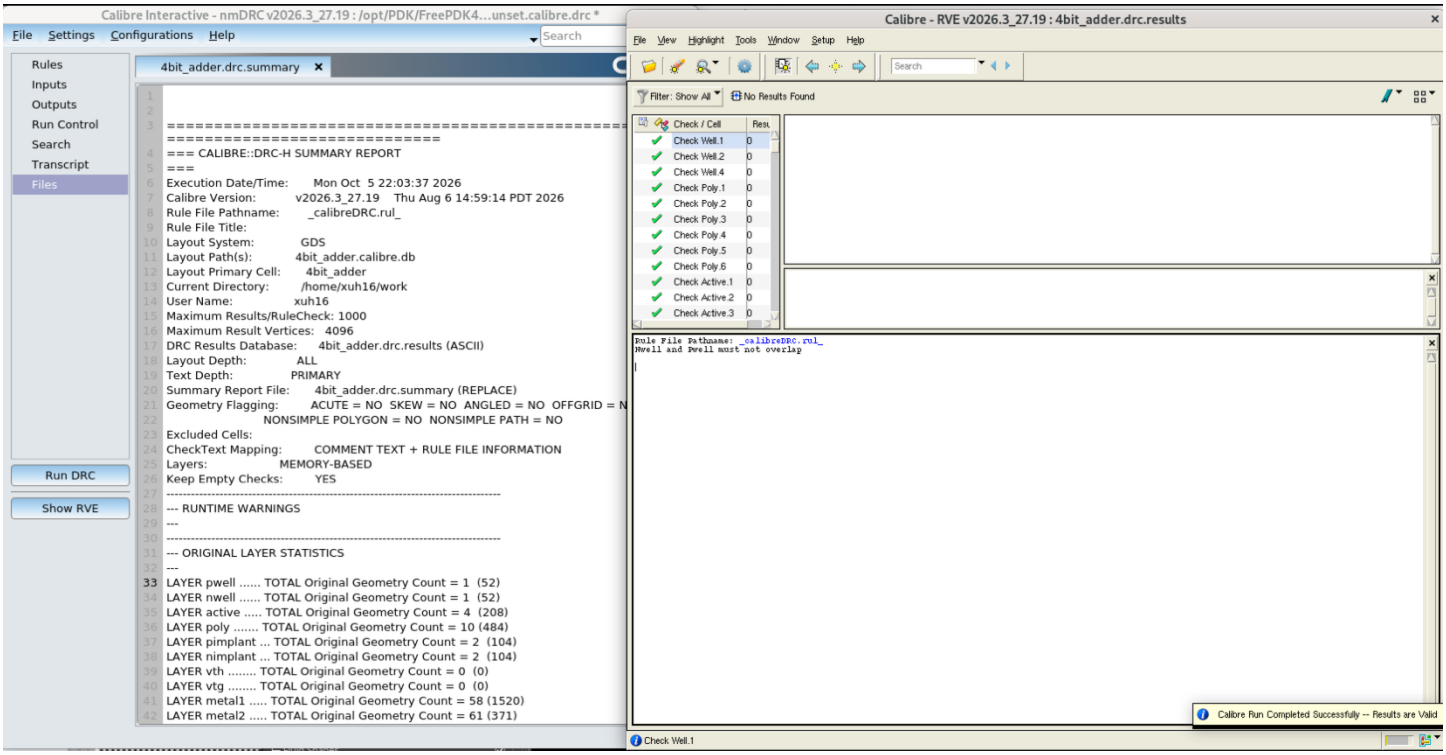
Simulation Results



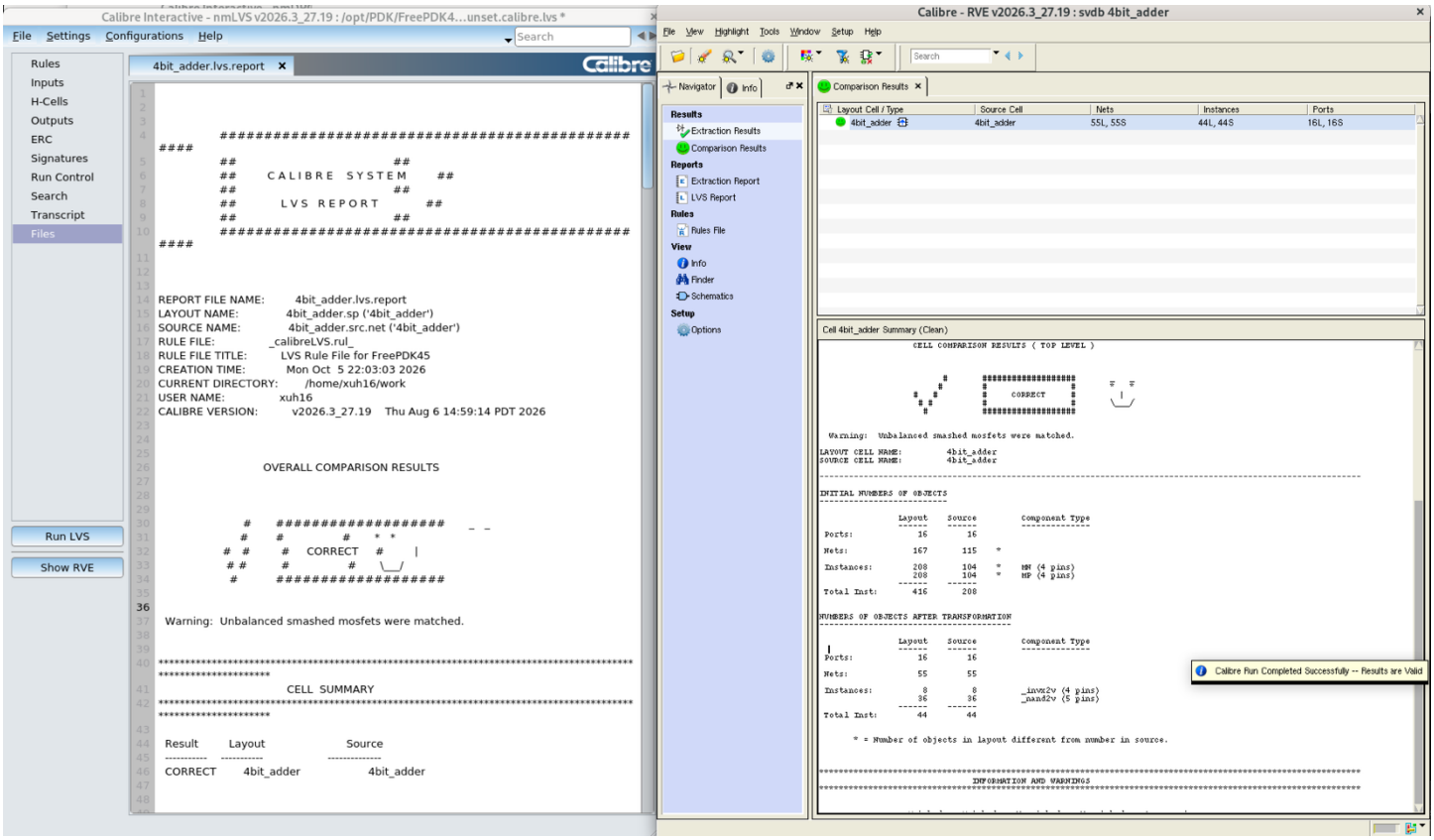
Layout



DRC


Calibre Interactive - nmDRC v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.drc *
File Settings Configurations Help
Rules
Inputs
Outputs
Run Control
Search
Transcript
Files
Run DRC
Show RVE
4bit_adder.drc.summary
=== CALIBRE::DRC-H SUMMARY REPORT
Execution Date/Time: Mon Oct 5 22:03:37 2026
Calibre Version: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
Rule File Pathname: _calibreDRC.rul_
Layout System: GDS
Layout Path(s): 4bit_adder.calibre.db
Layout Primary Cell: 4bit_adder
Current Directory: /home/xuh16/work
User Name: xuh16
Maximum Results/RuleCheck: 1000
Maximum Result Vertices: 4096
DRC Results Database: 4bit_adder.drc.results (ASCII)
Layout Depth: ALL
Text Depth: PRIMARY
Summary Report File: 4bit_adder.drc.summary (REPLACE)
Geometry Flagging: ACUTE = NO SKEW = NO ANGLED = NO OFFGRID = N
NONSIMPLE POLYGON = NO NONSIMPLE PATH = NO
Excluded Cells:
CheckText Mapping: COMMENT TEXT + RULE FILE INFORMATION
Layers: MEMORY-BASED
Keep Empty Checks: YES
--- RUNTIME WARNINGS
--- ORIGINAL LAYER STATISTICS
LAYER pwell ..... TOTAL Original Geometry Count = 1 (52)
LAYER nwell ..... TOTAL Original Geometry Count = 1 (52)
LAYER active ..... TOTAL Original Geometry Count = 4 (208)
LAYER poly ..... TOTAL Original Geometry Count = 10 (484)
LAYER pimplant ... TOTAL Original Geometry Count = 2 (104)
LAYER nimplant ... TOTAL Original Geometry Count = 2 (104)
LAYER vth ..... TOTAL Original Geometry Count = 0 (0)
LAYER vtg ..... TOTAL Original Geometry Count = 0 (0)
LAYER metal1 ..... TOTAL Original Geometry Count = 58 (1520)
LAYER metal2 ..... TOTAL Original Geometry Count = 61 (371)
Calibre - RVE v2026.3_27.19 : 4bit_adder.drc.results
File View Highlight Tools Window Setup Help
Filter: Show All
No Results Found
Check / Cell
Check Well.1
Check Well.2
Check Well.4
Check Poly.1
Check Poly.2
Check Poly.3
Check Poly.4
Check Poly.5
Check Poly.6
Check Active.1
Check Active.2
Check Active.3
Rule File Pathname: _calibreDRC.rul_
Nwell and Pwell must not overlap
Calibre Run Completed Successfully -- Results are Valid
Check Well.1

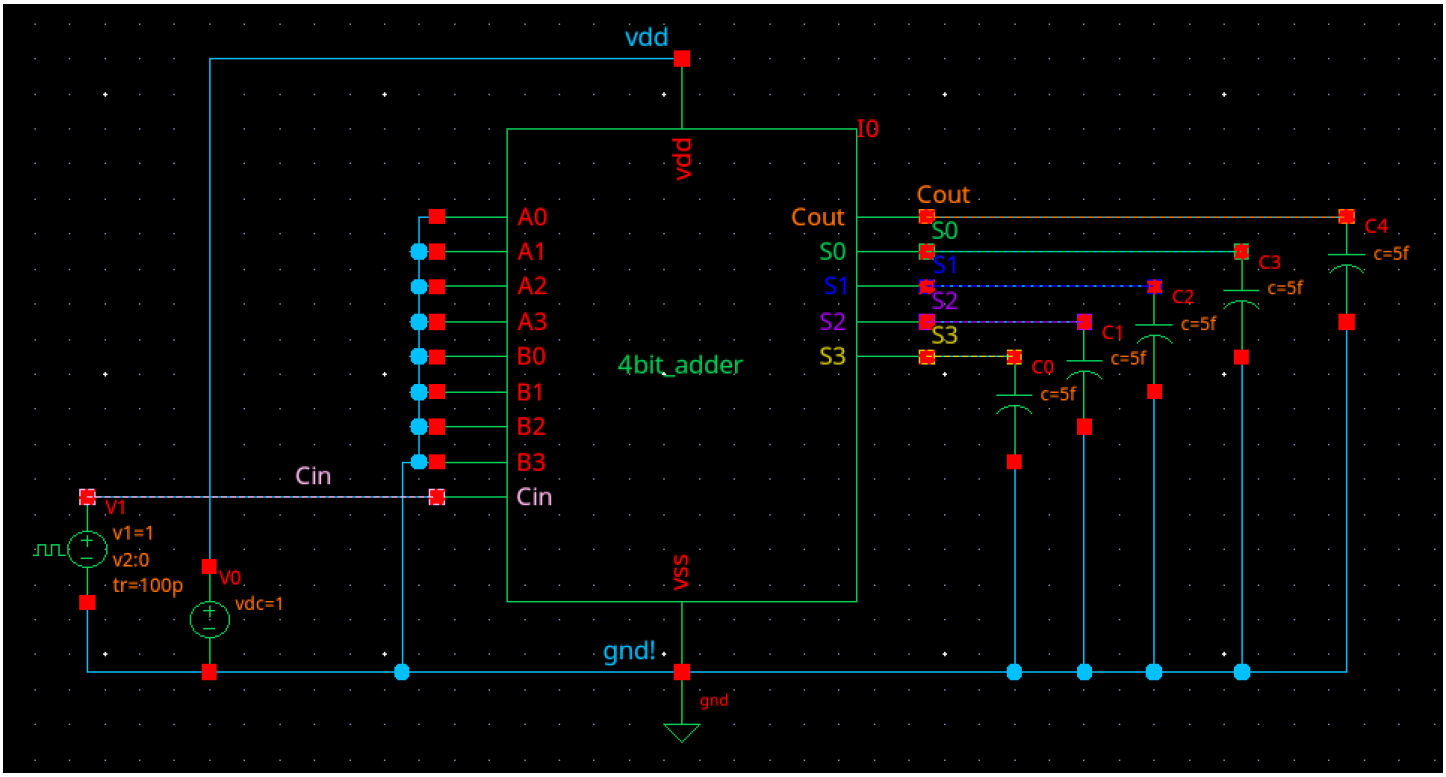
LVS


Calibre Interactive - nmLVS v2026.3_27.19 : /opt/PDK/FreePDK4...unset.calibre.lvs *
File Settings Configurations Help
Rules
Inputs
H-Cells
Outputs
ERC
Signatures
Run Control
Search
Transcript
Files
Run LVS
Show RVE
4bit_adder.lvs.report
LVS REPORT
REPORT FILE NAME: 4bit_adder.lvs.report
LAYOUT NAME: 4bit_adder.sp ('4bit_adder')
SOURCE NAME: 4bit_adder.src.net ('4bit_adder')
RULE FILE: _calibreLVS.rul_
RULE FILE TITLE: LVS Rule File for FreePDK45
CREATION TIME: Mon Oct 5 22:03:03 2026
CURRENT DIRECTORY: /home/xuh16/work
USER NAME: xuh16
CALIBRE VERSION: v2026.3_27.19 Thu Aug 6 14:59:14 PDT 2026
OVERALL COMPARISON RESULTS
CORRECT
Warning: Unbalanced smashed mosfets were matched.
CELL SUMMARY
Result Layout Source
CORRECT 4bit_adder 4bit_adder
Calibre - RVE v2026.3_27.19 : svdb 4bit_adder
File View Highlight Tools Window Setup Help
Navigator
Info
Results
Extraction Results
Comparison Results
Reports
Extraction Report
LVS Report
Rules
Rules File
View
Info
Finder
Schematics
Setup
Options
Comparison Results
Layout Cell / Type
Source Cell
Nets
Instances
Ports
4bit_adder
Cell 4bit_adder Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
Warning: Unbalanced smashed mosfets were matched.
LAYOUT CELL NAME: 4bit_adder
SOURCE CELL NAME: 4bit_adder
INITIAL NUMBERS OF OBJECTS
Component Type
Ports: 16 16
Nets: 167 115 *
Instances: 208 104 * NM (4 pins)
208 104 * MP (4 pins)
Total Inst: 416 208
NUMBERS OF OBJECTS AFTER TRANSFORMATION
Ports: 16 16
Nets: 55 55
Instances: 8 8 _invx2v (4 pins)
36 36 _nand2v (5 pins)
Total Inst: 44 44
* = Number of objects in layout different from number in source.
INFORMATION AND WARNINGS
Calibre Run Completed Successfully -- Results are Valid

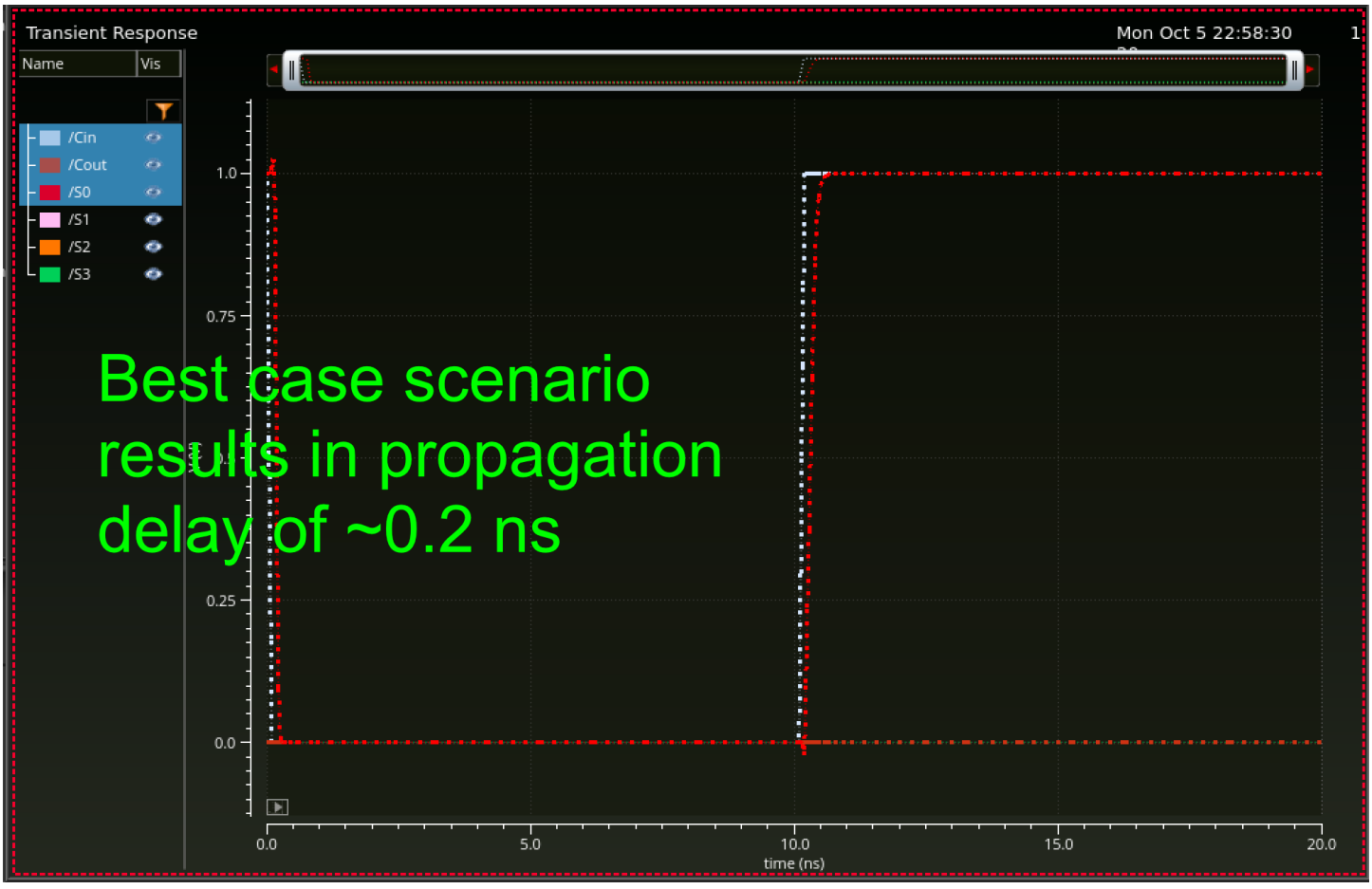
4-Bit Adder Critical Path Analysis

To test the critical path propagation delays, I set up two more simulations. One was the “best case,” where A=0000, B=0000, and Cin changes from 0 to 1. This is the best case because the carry does not have to propagate, and only S0 updates. One was the “worst case,” where A=1111, B=0000, and Cin changes from 0 to 1. This is the worst case because the carry has to ripple through all four full adders. The worst case resulted in a propagation delay 5x that of the best case (1ns vs 0.2ns). Results are shown below.

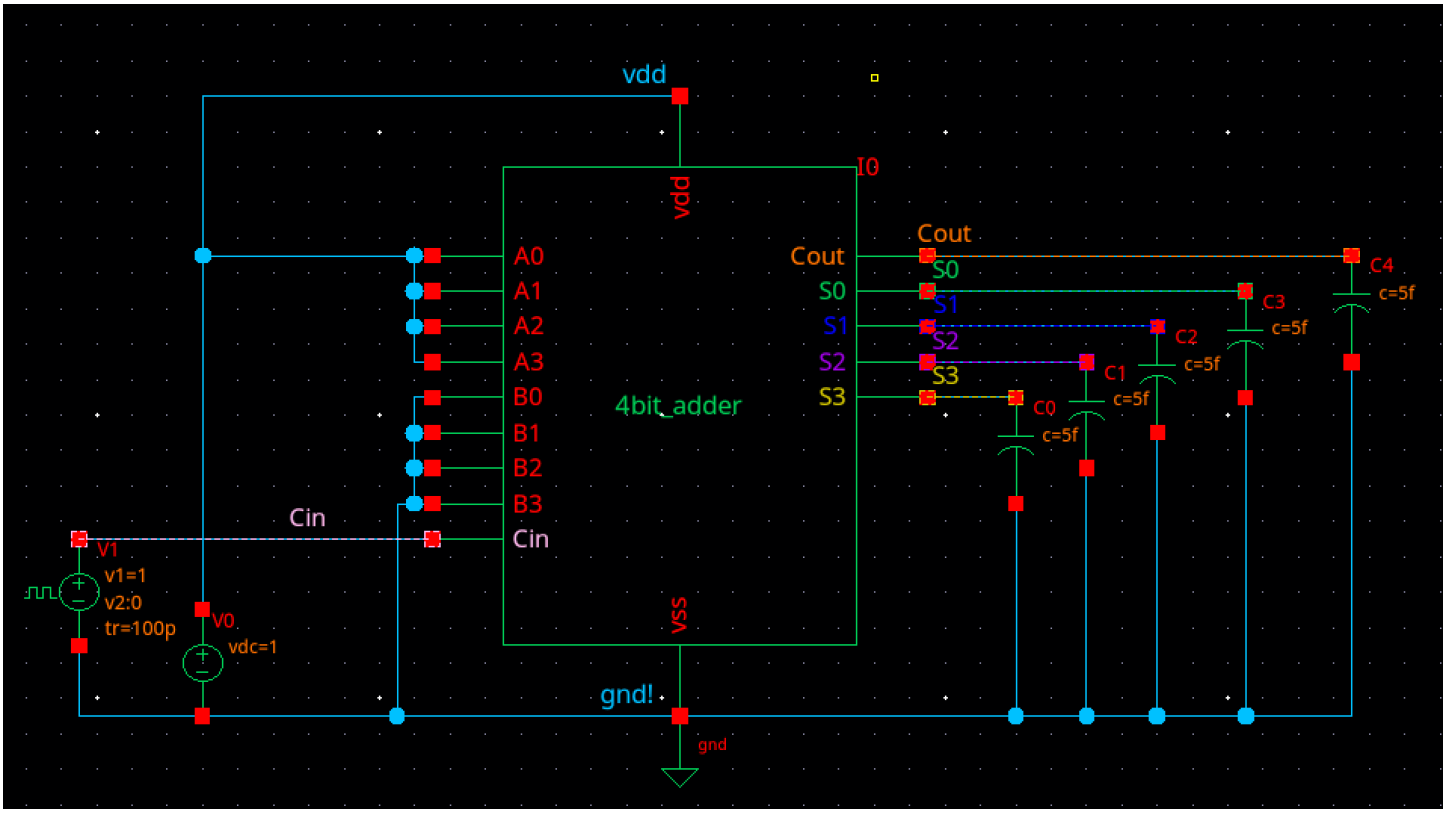
Best case setup



Best case results



Worst case setup



Worst case results

